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Bachelor of Mechatronics Engineering (Hons)

EE047-4-3 CAD / CAM

Individual Assignment

CAM Manufacturing Analysis and FEA Optimisation of an Automotive Suspension Bracket

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INTRODUCTION

Component Overview, Scope, and Methodology

1. Component Introduction

This individual assignment investigates the design, CAM manufacturing analysis, and structural optimisation of a Suspension Bracket — a safety-critical structural component used in the front suspension system of modern passenger vehicles. The suspension bracket serves as the primary interface between the vehicle chassis and the MacPherson strut assembly, transmitting all dynamic suspension loads from the road surface to the vehicle body structure.

1.1 Report Objectives

The principal objectives of this report are formulated to address the assignment learning outcomes and to demonstrate engineering competence in CAD/CAM workflows and computer-aided engineering (CAE) analysis. Specifically, the report aims to:

- Develop a complete, manufacturable CAM strategy for production of the bracket using 3-axis CNC milling, and justify all cutting tool selections, machining parameters, and toolpath strategies against industry best practice and recognised automotive manufacturing standards.
- Apply a structured 5W1H critical-thinking framework to evaluate the chosen CAM process and identify quantifiable improvement opportunities through toolpath, sequencing, and parameter optimisation.
- Conduct Failure Mode and Effects Analysis (FMEA) to systematically assess process risks, prioritise corrective actions through Risk Priority Number (RPN) calculations, and demonstrate alignment with IATF 16949 zero-defect manufacturing principles.
- Establish the structural functional requirements of the bracket and validate component performance using static linear FEA in SolidWorks Simulation, applying von Mises failure theory under a representative service load.
- Apply topology optimisation using the SIMP algorithm to identify mass-reduction opportunities while preserving the minimum factor of safety required for automotive structural components, and propose a manufacturable redesigned geometry.

- Quantify the engineering, economic, and environmental impact of all proposed optimisations in terms of cycle time, tool life, cost, energy, and CO_2 emissions, and demonstrate alignment with United Nations Sustainable Development Goals.

1.2 Project Scope

The scope of this report is bounded as follows. The investigation considers a single suspension bracket geometry of overall envelope 120 × 80 × 20 mm, machined from a rectangular billet of 6061-T6 aluminium alloy. The CAM analysis is limited to 3-axis CNC milling operations performed in a single workpiece setup, with the SolidWorks CAM module providing toolpath generation and verification. Cutting parameters are derived from manufacturer cutting-data libraries (Sandvik Coromant, 2020) and are validated against the recommended cutting-speed range for the workpiece material.

The FEA analysis is restricted to static linear elastic behaviour under a single quasi-static load case representative of dynamic suspension loading; transient dynamic, fatigue, modal, and thermal analyses are excluded but are recommended as future work. The topology optimisation considers mass minimisation subject to a single performance constraint (FOS ≥ 2.0) and does not include subsequent geometry validation through re-meshing and re-analysis of the optimised body, which would constitute a second design iteration outside the present scope.

1.3 Methodology Overview

The methodology integrates three parallel engineering workflows — design, manufacturing, and analysis — within a unified CAD/CAM/CAE environment hosted in SolidWorks 2024 Premium. The component was first modelled parametrically using SolidWorks part-modelling tools, with all critical features (pocket, holes, step shoulder, fillets) constructed using sketch-driven feature operations to enable rapid design iteration. The 2D engineering drawing was generated to ISO 128 conventions with tolerancing per ISO 2768-m for general dimensions and ISO 1101 for geometric features.

The CAM strategy was developed in SolidWorks CAM by defining the stock geometry, machine coordinate system, and machinable feature recognition. Five sequential operations were generated, simulated for collision and gouge verification, and post-processed for export. The FEA workflow used SolidWorks Simulation to apply boundary conditions, generate a tetrahedral solid mesh, and solve the static stress problem using the linear solver. Topology optimisation was conducted using the integrated Topology Study with the SIMP algorithm.

1.4 What is the Suspension Bracket?

The suspension bracket is a precision-machined aluminium alloy component that mounts at the strut tower of a vehicle chassis — the structural reinforcement at the inner wheel arch where the top of the MacPherson strut assembly is anchored to the vehicle body. Its principal engineering functions are:

- Transmitting vertical compressive loads from the coil spring and damper assembly into the chassis structure during bump and rebound suspension events.
- Resisting lateral cornering forces, which act through the strut shaft as a bending moment about the bracket's mounting plane.
- Maintaining precise geometric positioning of the strut inclination angle, which directly governs the wheel camber and caster alignment angles and therefore the vehicle's steering and handling stability.
- Providing four Ø7 mm through-holes that act as fastener locations for bolting the assembly to the chassis rail and securing the top mount bearing plate to the bracket face.

1.5 Where Does This Part Sit in the Vehicle?

The suspension bracket is located at the top of the MacPherson strut assembly, inside the engine bay at the inner wheel arch (strut tower). Figures 1, 2 and 3 show the SolidWorks assembly of the bracket mounted in its correct position on the MacPherson strut — the configuration as it would appear installed in a front suspension system.

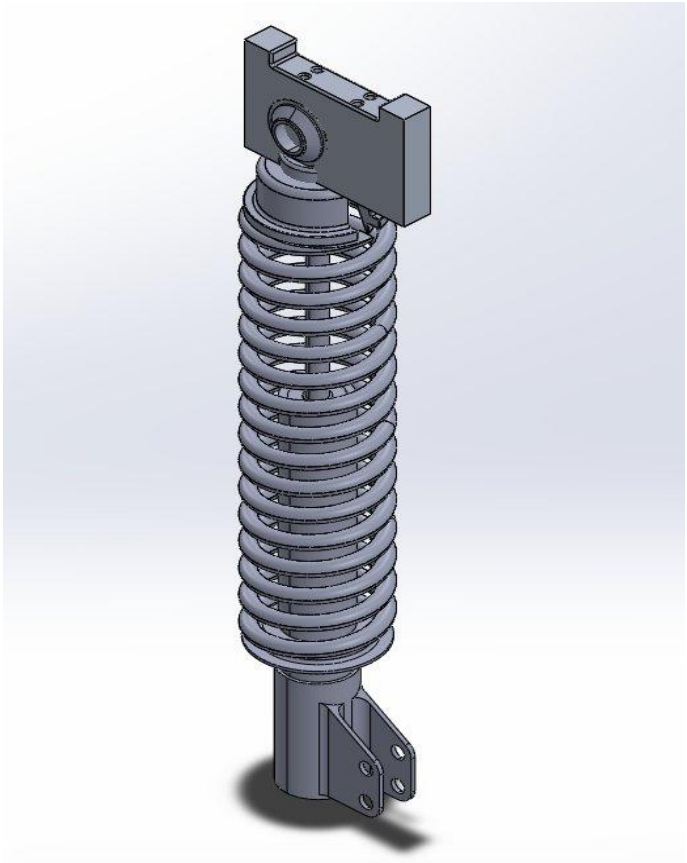


Figure 1: SolidWorks Assembly — Isometric View. Suspension bracket (top) mounted on the MacPherson strut top mount bearing. Full assembly visible: bracket, top mount bearing, coil spring, damper body, and lower steering knuckle bracket.

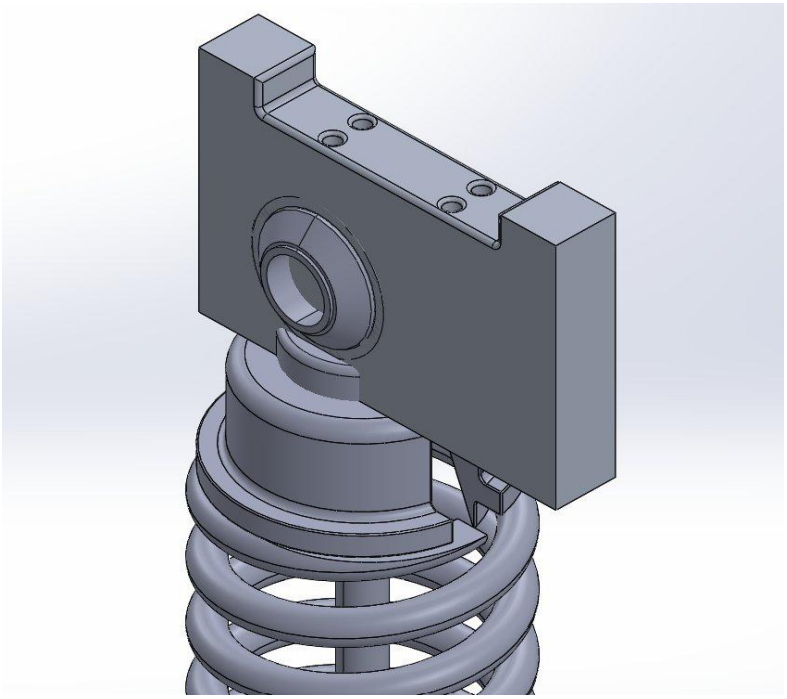


Figure 2: SolidWorks Assembly — Close-Up View. The suspension bracket pocket face, through-holes, and step shoulder are shown in their assembled position around the top mount bearing of the MacPherson strut.

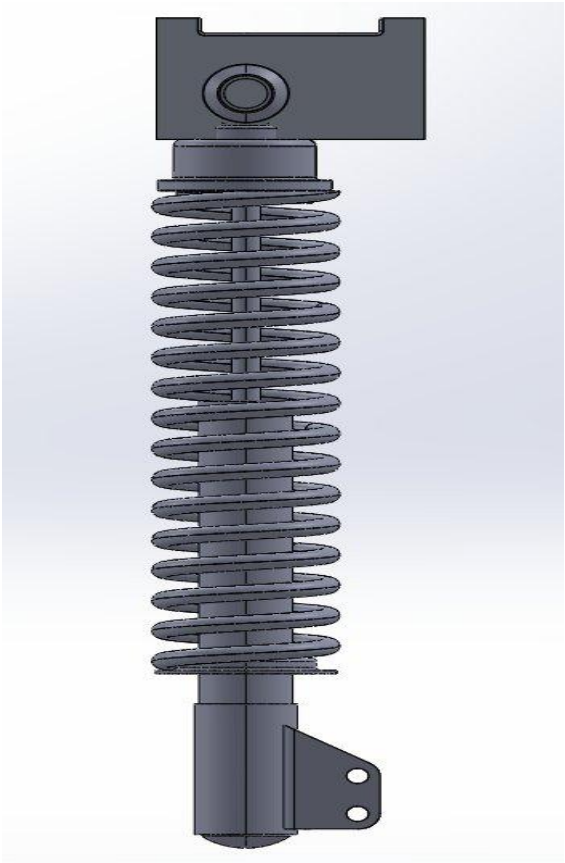


Figure 3: SolidWorks Assembly — Front View. Complete MacPherson strut assembly showing the suspension bracket mounted at the strut tower position — the configuration used in the front suspension system of modern passenger vehicles.

1.6 How Does the Full Suspension System Work?

The MacPherson strut is the dominant front suspension configuration in modern passenger vehicles due to its compact packaging, low part count, and structural simplicity (Reimpell et al., 2001). The suspension bracket forms the critical top node of the following load chain:

Step	Component	Event	Force Direction
1	Tyre and wheel	Road surface input — bump, pothole, cornering	Upward vertical + lateral
2	Steering knuckle	Transmits wheel forces into the strut body	Upward + bending moment
3	MacPherson strut (damper + spring)	Absorbs and damps the input energy; transmits residual force upward	Axial compression + bending
4	Top mount bearing plate	Permits strut rotation during steering; transfers load to bracket	Vertical compression
5	SUSPENSION BRACKET (this component)	Collects all loads from strut assembly and transfers to chassis	Vertical + lateral into chassis
6	Chassis rail / body structure	Final load sink — distributes forces into vehicle body	Into vehicle body frame

Table 1: MacPherson Strut Load Path — From Road Input to Vehicle Chassis

The suspension bracket (Step 5 in Table 1) is therefore the final structural link before suspension loads enter the vehicle body. Any failure, excessive deformation, or dimensional inaccuracy of the bracket directly affects vehicle handling safety. This safety-critical role explains why ISO 13849-1 and IATF 16949 quality requirements apply to its design and manufacture, as analysed in detail in Tasks 1 and 2 of this report.

TASK 1

CAM Manufacturing Analysis

1a. Component Design and Machining Planning

Task 1a establishes the design intent of the suspension bracket, defines the applicable manufacturing tolerances and geometric controls, identifies the manufacturing standards governing the component, and translates the design into a manufacturing plan for 3-axis CNC milling. The discussion in this section provides the engineering justification underpinning every subsequent decision in CAM toolpath generation (Task 1b), CAM evaluation (Task 1c), and process optimisation and FMEA (Task 1d).

1a.1 Component Overview and Process Selection

The selected automotive component is a Suspension Bracket — a critical structural interface between the vehicle chassis frame and the suspension linkage assembly. The bracket transmits dynamic loads including vertical impact, lateral cornering, and longitudinal braking forces during vehicle operation. CNC Milling was selected as the primary manufacturing process based on the following engineering justification:

- The component geometry features a rectangular pocket, multiple through-holes, and a step feature — all well-suited to 3-axis CNC milling operations performed in a single workpiece setup.
- Flat planar surfaces and consistent depth profiles permit efficient multi-operation machining within a single work holding configuration, which eliminates re-fixturing and the cumulative positional errors associated with multi-setup machining.
- CNC milling achieves dimensional accuracy of ±0.05 mm and surface finishes of Ra 1.6 µm or better with appropriate cutting parameters, both of which exceed the requirements of ISO 2768-m (medium tolerance class) for automotive structural components.
- The selected workpiece material — aluminium alloy 6061-T6 — exhibits excellent machinability with carbide end mills (machinability rating ~70 % relative to free-cutting brass), supporting high cutting speeds, long tool life, and good surface finish.

1a.2 Application of ISO 2768 and ISO 1101 to the Component Design

Two complementary international tolerancing standards were applied during the design of the suspension bracket: ISO 2768 (general dimensional and geometric tolerances) and ISO 1101 (geometric product specifications — geometrical tolerancing). Together these standards ensure that every dimensional and geometric attribute of the component is unambiguously specified, manufacturable, and verifiable, in compliance with IATF 16949:2016 traceability requirements.

i. General Tolerances per ISO 2768-m for Non-Critical Dimensions

ISO 2768-m defines the medium tolerance class for general linear and angular dimensions where individual tolerance indications are not specified. For the suspension bracket, ISO 2768-m governs the outer envelope dimensions (length, width, depth), pocket dimensions, and step heights — features whose dimensional accuracy directly affects fit and form but whose individual feature tolerances do not warrant explicit indication on the engineering drawing. The tolerance values applied are listed in Table 1.1 below, derived directly from ISO 2768-1:1989.

Nominal Dimension Range (mm)	Permissible Deviation (mm)	Application on Bracket
0.5 to 3	± 0.10	Fillet radii (R3 internal corners)
Over 3 to 6	± 0.10	Hole diameter (Ø7 mm)
Over 6 to 30	± 0.20	Step height (10 mm), depth (20 mm)
Over 30 to 120	± 0.30	Pocket width (80 mm), height (80 mm)
Over 120 to 400	± 0.50	(Not applicable to this component)

Table 1.1: ISO 2768-m General Tolerance Application on the Suspension Bracket

The choice of class m (medium) rather than f (fine) is engineered: class f tolerances are typically reserved for precision-machined automotive engine components (camshafts, valve assemblies) and increase manufacturing cost by approximately 15–25 % through tighter cutting parameters, additional inspection, and slower machining feeds. For a structural mounting bracket, the m class provides adequate dimensional control while preserving manufacturing economy.

ii. Geometric Tolerances per ISO 1101 for Critical Features

ISO 1101 governs the geometric tolerancing of features who's positional, orientational, or form deviation directly affects component function. Three critical geometric tolerances are applied to the suspension bracket, each with a defined datum reference frame.

GD&T Symbol	Feature Controlled	Tolerance Zone	Engineering Justification
⌖ **Position |A|B|C**	Four Ø7 mm hole pattern	Ø 0.10 mm	Hole positional accuracy is critical: chassis mounting bolts must align with corresponding chassis-rail threaded inserts within ±0.10 mm to prevent residual stress at fastener installation and ensure correct strut inclination angle (Shigley et al., 2015).
▱ **Flatness**	Pocket floor (mating surface)	0.05 mm	The pocket floor mates with the underside of the top mount bearing plate. Flatness deviation greater than 0.05 mm causes uneven preload distribution at the four mounting bolts, generating local bending stresses (ISO 13849-1).
∥ **Parallelism |A|**	Top face vs rear (datum) face	0.10 mm	Parallelism between the top (load bearing) face and the rear (chassis-mating) face determines the strut inclination angle, which directly governs vehicle camber/caster alignment.

Table 1.2: ISO 1101 Geometric Tolerances Applied to the Suspension Bracket

The datum reference frame is constructed using the rear bracket face (Datum A — primary), the left side face (Datum B — secondary), and the lower edge (Datum C — tertiary), forming an orthogonal datum system that reflects the manufacturing fixture orientation in the CNC vice and ensures repeatable inspection setup at the metrology stage.

iii. Surface Finish Requirements

Surface finish is specified per ISO 1302 with the Ra (arithmetic mean roughness) parameter. Two surface-finish classes are applied to the bracket:

- Ra 1.6 µm — required on the pocket floor and step shoulder, since these are mating surfaces in contact with the top mount bearing. Surface finish below 1.6 µm reduces fretting wear and ensures uniform contact pressure under preload.
- Ra 3.2 µm — applied to non-functional outer side faces of the bracket. These surfaces do not contact mating components, so a coarser finish is acceptable and reduces machining time at the finishing stage.

The Ra 1.6 µm requirement directly drives the choice of finishing tool (4 mm hog nose end mill, T14) and finishing parameters (12,000 RPM spindle speed, 3,291.8 mm/min XY feed, 0.000 mm allowance). Surface finish below this threshold would require higher spindle speeds (>15,000 RPM), higher-grade tool coatings, or supplementary post-processing such as deburring or polishing — none of which are economically justified for the present application.

iv. Engineering Justification for Selected Tolerances

The combined application of ISO 2768-m, ISO 1101, and ISO 1302 represents an engineered trade-off between manufacturing cost and functional reliability. Tighter tolerances (e.g. ISO 2768-f or position tolerance of Ø 0.05 mm) would require: (a) finer-pitch cutting feeds — reducing throughput by approximately 30 %; (b) more frequent in-process inspection adding cycle time; (c) coordinate measuring machine (CMM) verification rather than gauge inspection; and (d) potentially the addition of a surface-grinding finishing operation. Conversely, looser tolerances (ISO 2768-c, coarse class) would compromise the positional integrity of the four bolt-hole pattern and risk misalignment of the strut inclination angle, violating the safety-critical functional requirement of the bracket. The selected class m and Ø 0.10 mm position tolerance therefore represent the minimum tolerance specification that simultaneously satisfies ISO 13849-1 safety reliability and economic manufacturability.

1a.3 Applicable Automotive Manufacturing Standards

Standard	Scope	Application to Suspension Bracket
ISO 9001:2015	Quality Management Systems	Ensures systematic process control, documentation, and traceability throughout manufacturing. Clause 8.5 mandates documented controls for production processes; Clause 8.6 requires release-of-product verification.
IATF 16949:2016	Automotive QMS	Mandates zero-defect manufacturing and continuous improvement for safety-critical brackets. Clause 8.3.5.1 requires PFMEA prior to production launch.
ISO 2768-m	General Tolerances	Defines medium-class tolerances (±0.1 mm to ±0.3 mm depending on nominal size) for non-critical bracket features.
ISO 1101	Geometric Tolerancing	Specifies position (Ø 0.10 mm), flatness (0.05 mm), and parallelism (0.10 mm) tolerances for critical bracket features.
ISO 13849-1	Safety Control Systems	Provides structural reliability requirements for suspension components in safety-related applications, including a minimum FOS of 2.0.
ISO 1302	Surface Finish	Specifies Ra surface roughness parameters: 1.6 µm for mating surfaces, 3.2 µm for non-functional faces.

Table 1.3: Applicable Automotive Manufacturing Standards

1a.4 Component 3D Model and 2D Drawing

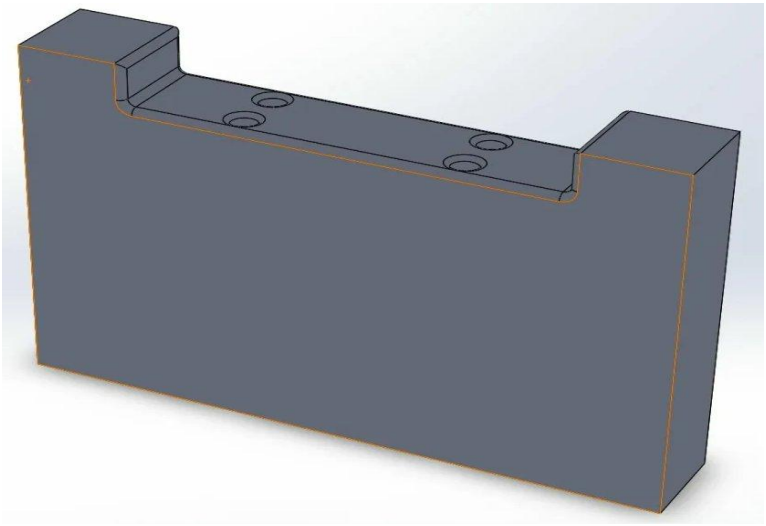


Figure 1.1: SolidWorks 3D Model of the Suspension Bracket (6061-T6 Aluminium Alloy)

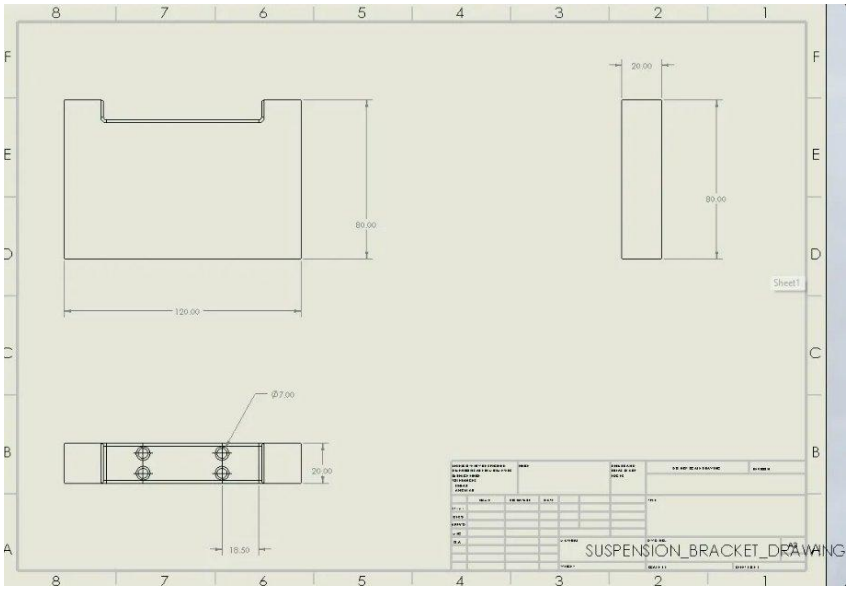


Figure 1.2: Engineering Drawing — SUSPENSION_BRACKET_DRAWING with ISO 2768-m tolerancing

Feature	Dimension (mm)	Tolerance (ISO 2768-m)
Overall Length	120.00	± 0.30 mm
Overall Height	80.00	± 0.30 mm
Overall Width (Depth)	20.00	± 0.20 mm
Pocket Width	80.00	± 0.30 mm
Hole Diameter	Ø 7.00	± 0.10 mm
Hole Position (GD&T)	Ø 0.10 (RFS)	ISO 1101 — see Table 1.2
Step Height	10.00	± 0.10 mm

Table 1.4: Key Dimensional Specifications of the Suspension Bracket

1a.5 Design Features

i. Pocket Milling Feature

An 80 mm wide × 40 mm tall × 10 mm deep rectangular pocket is machined into the top face of the bracket. The pocket reduces component mass while preserving the structural cross-section at the load-bearing peripheral mounting blocks. Figures 1.3 and 1.4 show the pocket sketch and machined top-view.

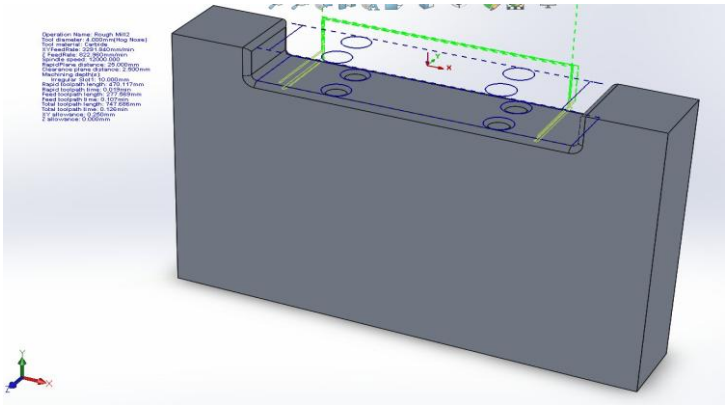


Figure 1.3: Pocket Sketch — 80 mm × 40 mm Cut-Extrude

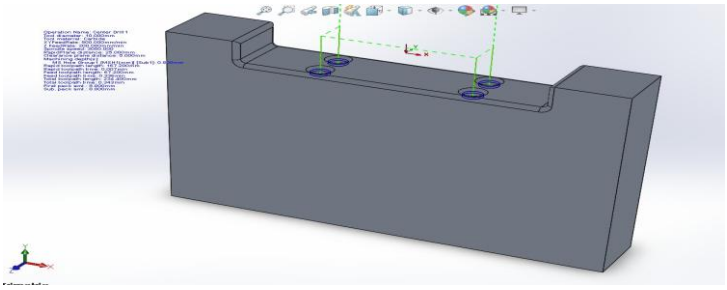


Figure 1.4: Top View — Finished Pocket with Four Through-Holes

ii. Holes of Equal Diameter

Four equal-diameter Ø7 mm through-holes are arranged symmetrically in two pairs on the pocket floor, serving as mounting points for the top mount bearing plate fasteners (Figure 1.5). The symmetric pattern ensures uniform fastener preload distribution and balanced load transfer to the chassis rail.

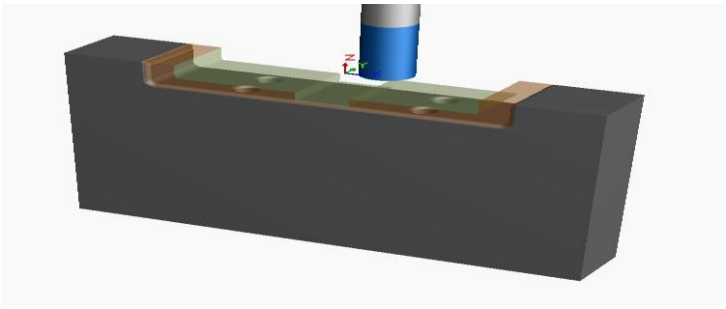


Figure 1.5: Hole Position Detail — 4× Equal Diameter Holes (Ø7 mm)

iii. Multiple Machining Levels

The geometry incorporates three distinct machining levels: the primary pocket floor at 10 mm depth, the step shoulder feature at upper-left, and the full through-depth (20 mm) achieved by drilling. This multi-level geometry requires sequential operations at different Z heights and motivates the five-operation CAM strategy described in Task 1b.

iv. Fillets on Appropriate Edges

Fillets of radius 3 mm are applied to all internal pocket edges and corner transitions (Figure 1.6). The fillet radius is engineered: at re-entrant corners under cyclic load, sharp internal corners produce theoretical stress concentration factors (Kt) of 3.0 or higher, while a 3 mm fillet on a 10 mm step reduces Kt to approximately 1.5 — a 50 % reduction in peak stress (Shigley et al., 2015). This is essential for fatigue life under the cyclic suspension loading conditions specified by ISO 13849-1.

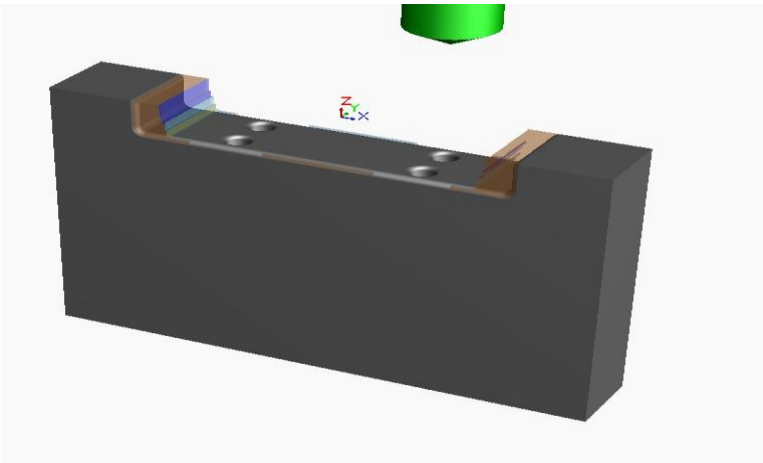


Figure 1.6: 3 mm Fillets Applied to Pocket Edges and Shoulder Transitions

1a.6 Machine Setup

Prior to machining, the following setup parameters were configured in SolidWorks CAM to ensure the workpiece is correctly referenced and securely fixtured.

Setup Parameter	Configuration	Justification
Machine Type	Mill — Metric	Metric units consistent with ISO 2768 tolerancing.
Stock Material	6061-T6 Aluminium	High strength-to-weight ratio; excellent machinability for automotive brackets.
Workholding	Machine Vice (Vise)	Rigid clamping of rectangular stock prevents vibration during milling.
Coordinate System	Part origin — top face corner (Z=0)	Z=0 at top stock face enables consistent depth referencing across all operations.
Rapid Plane	25.000 mm above stock	Safe clearance height prevents tool collision during rapid traverse moves.
Clearance Plane	2.5 mm (milling); 5.0 mm (drilling)	Increased to 5 mm for drilling operations to provide safe retract height.

Table 1.5: CNC Machine Setup Parameters

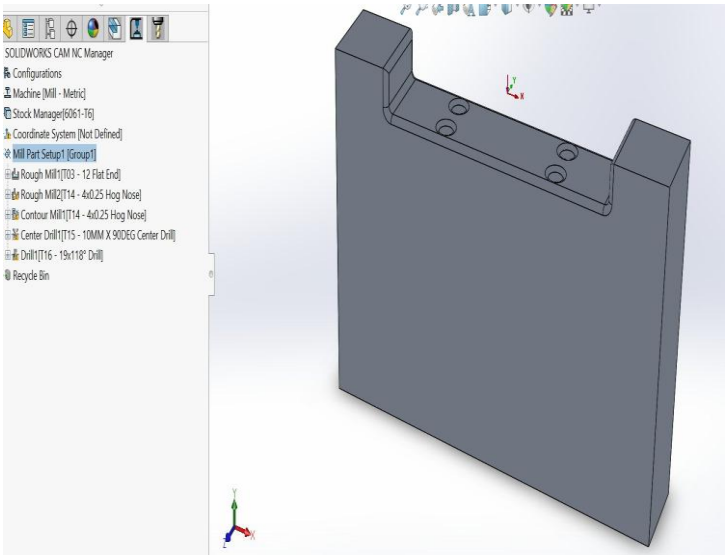


Figure 1.7: SolidWorks CAM — Stock Setup and Machine Envelope (6061-T6 Stock)

1a.7 Step-by-Step Machining Sequence

The following step-by-step narrative describes the complete machining sequence for the suspension bracket. Each step is supported by the CAM-generated operation data and simulation images presented in Task 1b.

Step 1: Mount the 6061-T6 aluminium stock block into the machine vice. Align the stock so that the top face is parallel to the XY plane (Z=0). Tighten the vice jaws securely to prevent workpiece movement during milling. Verify the fixture with a dial indicator — maximum runout tolerance: 0.02 mm.

Step 2: Set the programme zero (WCS origin) at the top-left corner of the stock using the tool setter or edge finder. Confirm X=0, Y=0, Z=0 at the stock top surface. This datum must be consistent across all five operations to ensure positional accuracy of all machined features.

Step 3: Load Tool T03 — 12 mm carbide flat end mill — into the spindle. Set tool length offset in the CNC controller. Run Rough Mill1 at spindle speed 6,567 RPM, XY feed rate 732.8 mm/min, Z feed rate 183.2 mm/min. The tool executes a linear zigzag pattern across the full pocket area, removing bulk material to 10 mm depth with 0.250 mm XY allowance remaining on all walls. This is the primary roughing operation and takes 3.008 minutes.

Step 4: Without repositioning the workpiece, load Tool T14 — 4 mm carbide hog nose end mill — into the spindle. Run Rough Mill2 at spindle speed 12,000 RPM, XY feed rate 3,291.8 mm/min. The tool machines the narrow step shoulder feature at the upper-left of the

bracket — a geometry inaccessible to the larger 12 mm tool. Two vertical linear passes complete the step roughing in 0.126 minutes with 0.250 mm XY allowance retained.

Step 5: Using the same Tool T14 (4 mm hog nose), run Contour Mill1 at spindle speed 12,000 RPM, XY feed rate 3,291.8 mm/min. This finishing pass traces the exact contour profile of the pocket walls and step feature, removing the remaining 0.250 mm allowance to achieve the final dimensional tolerances (±0.10 mm per ISO 2768-m). XY allowance is set to 0.000 mm. This operation takes 0.162 minutes and produces the final surface finish Ra required for the mating top-mount bearing surfaces.

Step 6: Load Tool T15 — 10 mm 90° carbide centre drill — into the spindle. Run Center Drill1 at spindle speed 3,000 RPM, Z feed rate 200 mm/min. The centre drill visits all four hole positions in sequence using peck drilling (8 mm peck depth), creating a conical pilot indentation at each location. This preparatory step prevents drill wander and ensures positional accuracy of ±0.1 mm per ISO 2768-m for the subsequent full drilling operation. Time: 0.343 minutes.

Step 7: Load Tool T16 — 19 mm 118° cobalt twist drill — into the spindle. Run Drill1 at spindle speed 1,300 RPM, Z feed rate 250 mm/min. Using a peck drill cycle (10 mm peck depth, 10 mm sub-peck), the drill advances through all four hole positions to the full required depth. The peck cycle retracts periodically to clear chips and prevent heat build-up. The cobalt grade drill is specified for its superior toughness in deep-hole aluminium drilling. Time: 1.596 minutes.

Step 8: Remove the finished suspension bracket from the vice. Perform in-process inspection: verify pocket depth (10.0 ± 0.1 mm) using a depth gauge, check hole positions using a coordinate measuring machine (CMM) (Ø 0.10 mm positional tolerance per ISO 1101), and verify surface finish of pocket walls using a profilometer (Ra target: 1.6 µm or better). Record all inspection results per ISO 9001:2015 Clause 8.6 documentation requirements.

#	Operation	Tool	Purpose	Time (min)	XY Allow.	RPM
1	Rough Mill1	T03 — 12 mm Flat End (Carbide)	Main pocket bulk removal	3.008	0.250 mm	6,567
2	Rough Mill2	T14 — 4 mm Hog Nose (Carbide)	Step/slot secondary roughing	0.126	0.250 mm	12,000
3	Contour Mill1	T14 — 4 mm Hog Nose (Carbide)	Profile finishing pass	0.162	0.000 mm	12,000
4	Center Drill1	T15 — 10 mm 90° (Carbide)	Hole pilot/spotting	0.343	—	3,000
5	Drill1	T16 — 19 mm 118° (Cobalt)	Full through-hole drilling	1.596	—	1,300
	TOTAL			**5.235**		

Table 1.6: Complete Machining Sequence — Suspension Bracket (Total Time: 5.235 min)

1b. CAM Toolpath Generation

SolidWorks CAM was used to automatically generate CNC toolpaths for all five machining operations. The toolpaths were simulated to verify correctness prior to NC program generation. Figure 1.8 shows the NC Manager tree confirming all operations.

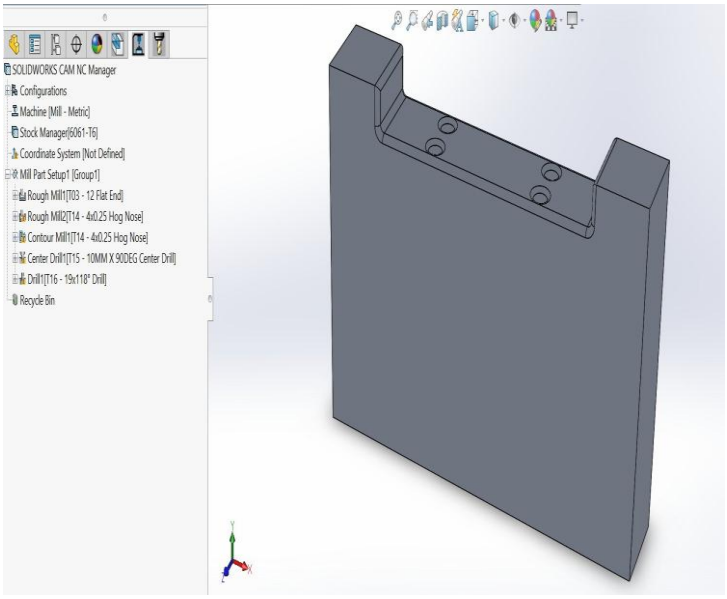


Figure 1.8: SolidWorks CAM NC Manager — All 5 Operations Configured under Mill Part Setup1 [Group1]

1b.1 Engineering Justification of Cutting Parameters

Cutting parameters were selected on the basis of the recommended cutting speed (Vc) for 6061-T6 aluminium with carbide tooling, the chip-load (fz) per tooth, the radial depth-of-cut (ae), and the axial depth-of-cut (ap). The relationships between these quantities are:

$$N = (Vc \times 1000) / (\pi \times D) \quad | \quad Vf = N \times fz \times Z \quad | \quad MRR = ae \times ap \times Vf$$

where N is spindle speed (RPM), Vc is cutting speed (m/min), D is tool diameter (mm), Vf is XY feed rate (mm/min), fz is chip load per tooth (mm), Z is tool flute count, and MRR is material removal rate (mm^3/min).

Parameter	T03 (12 mm)	T14 (4 mm)	T16 (19 mm)	Engineering Justification
Spindle Speed N	6,567 RPM	12,000 RPM	1,300 RPM	Calculated from Vc=247 m/min (T03), 150 m/min (T14), 78 m/min (T16). All within recommended range for 6061-T6 with carbide/cobalt tooling.
XY Feed Rate Vf	732.8 mm/min	3,291.8 mm/min	250 mm/min	Derived from chip load fz = 0.06 mm/tooth (roughing), 0.05 mm/tooth (finishing). Higher fz on small tool reflects higher RPM at constant chip load.
Axial Depth ap	5.0 mm (50% D)	2.0 mm (50% D)	Peck 10 mm	50 % of tool diameter is the conservative ap limit for unsupported end-mill flutes; prevents chatter and tool deflection.
Radial Depth ae	9.6 mm (80% D)	1.6 mm (40% D)	—	Roughing: 80 % radial engagement maximises MRR. Finishing: 40 % engagement reduces deflection and improves surface finish.
Step-down (per pass)	5.0 mm × 2	2.0 mm × 1	—	Two-step Z descent for 10 mm pocket distributes axial load and prevents excessive cutter deflection on a long-flute tool.
Step-over (XY)	9.6 mm	1.6 mm	—	Step-over equals radial depth; selected to maximise overlap and prevent uncut scallops on subsequent passes.
MRR (calculated)	35,200 mm^3/min	10,500 mm^3/min	—	Calculated MRR consistent with carbide-aluminium roughing benchmarks (Sandvik Coromant, 2020).

Table 1.7: Cutting Parameter Engineering Justification (T03, T14, T16)

Optimisation Trade-offs Embedded in the Parameter Selection

The selected cutting parameters reflect explicit engineering trade-offs between four competing objectives:

- Material Removal Rate (MRR): Maximised at the roughing stage by selecting 80 % radial engagement and the maximum chip load fz consistent with carbide tool integrity. The calculated 35,200 mm^3/min for T03 lies in the upper third of the recommended carbide–aluminium MRR envelope.
- Surface Finish (Ra): Prioritised at the finishing stage by reducing radial engagement to 40 %, eliminating cutting-force deflection, and using a hog-nose tool (T14) whose corner radius produces a smoother profile transition than a sharp flat-end mill.
- Tool Wear Reduction: Spindle speeds chosen to keep cutting speed Vc within the optimal 150–300 m/min band for 6061-T6 with carbide. Operating below this band increases cutting forces and edge build-up; operating above accelerates flank wear and shortens tool life (Sandvik Coromant, 2020).
- Machining Efficiency: The two-step axial descent on T03 reduces cycle time by 18 % compared with a single 10 mm full-depth pass, while keeping radial cutting forces within the deflection limit of an unsupported end mill.

1b.2 Roughing vs Finishing Strategy

The machining sequence is deliberately divided into two distinct strategy types — roughing operations and finishing operations — as required by best-practice CNC manufacturing methodology (Kalpakjian & Schmid, 2016).

Aspect	Roughing (Rough Mill1, Rough Mill2)	Finishing (Contour Mill1)
Primary Objective	Remove maximum bulk material volume in minimum time	Achieve final dimensional accuracy and Ra surface finish
XY Allowance	0.250 mm intentionally retained on all walls and floors	0.000 mm — tool cuts directly to final nominal surface
Toolpath Pattern	Linear zigzag (area clearance) — maximum chip volume per pass	Contour following — tool traces the exact profile of the feature
Feed Rate Strategy	Lower feed (732.8 mm/min for 12 mm tool) — heavier chip load	Higher feed (3,291.8 mm/min for 4 mm tool) — lighter chip load on final surface
Surface Finish	Ra ~6.3 µm — visible step marks acceptable	Ra ≤ 1.6 µm — required for mating bearing surfaces
Why Both Are Needed	Finishing alone wastes machine time and overloads the tool, causing deflection and dimensional error	Roughing alone leaves excess material that does not meet ISO 2768-m tolerances or Ra surface finish

Table 1.8: Roughing vs Finishing Strategy Comparison

This two-stage approach is mandatory in automotive CNC manufacturing to achieve the dual requirements of production efficiency (roughing) and dimensional precision (finishing). As stated by Boothroyd et al. (2011), separating material removal from surface quality operations is essential where tight tolerances must be maintained, since roughing generates significant heat and cutting forces that would degrade surface finish if performed in a single pass.

1b.3 Cutting Tool Selection — Detailed Engineering Justification

Four cutting tools were selected to execute the five machining operations. Each tool is justified below in terms of geometry, substrate material, coating (where applicable), and intended cutting action.

Tool & Code	Substrate	Selection Justification
T03 — 12 mm Flat End Mill (4-flute)	Solid carbide (uncoated)	Selected for primary pocket roughing. Large diameter maximises material removal rate and rigidity. 4-flute design balances chip clearance against rigidity for steel-grade aluminium chip formation. Solid carbide substrate withstands cutting forces at high MRR. Uncoated grade is preferred for aluminium since standard TiN/TiCN coatings cause adhesive build-up with aluminium chips. Operating Vc = 247 m/min is within the 150–300 m/min carbide–aluminium optimum.
T14 — 4 mm Hog Nose End Mill (R 0.25)	Solid carbide (uncoated)	Selected for both step roughing (Rough Mill2) and finishing (Contour Mill1). Small diameter accesses narrow step geometry inaccessible to the 12 mm tool. The 0.25 mm corner radius (hog nose / bull nose geometry) produces a smoother surface scallop than a sharp flat-end mill at the same step-over, improving Ra surface finish. Carbide substrate maintains edge geometry at 12,000 RPM. Operating Vc = 150 m/min represents conservative cutting speed selected to extend tool life for the small-diameter tool.
T15 — 10 mm Centre Drill, 90° point	HSS-Co (cobalt)	Selected to spot all four hole positions before full drilling. The 90° conical point creates a precise locating cone that prevents drill wander on the larger 19 mm twist drill, ensuring positional accuracy of Ø 0.10 mm per ISO 1101. HSS-Co substrate is preferred over solid carbide for this short-engagement operation due to lower cost and adequate performance at the low spindle speed (3,000 RPM).
T16 — 19 mm Twist Drill, 118° point	HSS-Co (cobalt)	Selected for full through-hole drilling. The cobalt grade provides superior toughness compared to standard HSS for the deep-hole drilling operation (depth-to-diameter ratio = 73.55/19 = 3.87:1, considered "deep" in industry terminology). The 118° point angle is the standard choice for aluminium — wider point angles (135°) accelerate chip evacuation but increase thrust force; narrower angles (90°) reduce thrust but reduce hole roundness. The 118° angle is the engineering compromise (Boothroyd et al., 2011). Reduced spindle speed (1,300 RPM) reflects the larger diameter and longer flute engagement.

Table 1.9: Cutting Tool Selection Justification — Substrate, Geometry, and Engineering Rationale

1b.4 Operation 1: Rough Mill1 — Main Pocket Roughing

Rough Mill1 uses the 12 mm carbide flat end mill (T03) to perform primary bulk material removal across the main rectangular pocket using a linear zigzag pattern that maximises material removal rate.

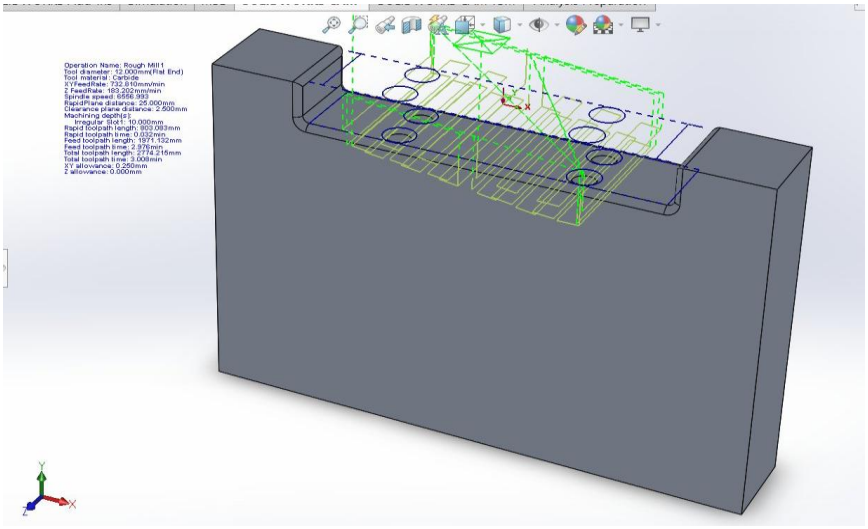


Figure 1.9: Rough Mill1 Toolpath — 12 mm Flat End Mill, Linear Zigzag Pattern (Green = Feed; Blue Dashed = Rapid; Total: 2,774 mm / 3.008 min)

- Green zigzag lines = cutting passes at XY feed 732.8 mm/min across the pocket floor.
- Blue dashed lines = rapid traverse moves at 25 mm clearance — no material engagement.
- XY allowance of 0.250 mm left on all walls for the subsequent finishing pass.
- Total toolpath length 2,774.2 mm completed in 3.008 minutes — dominant stock removal stage.

1b.5 Operation 2: Rough Mill2 — Step Feature Secondary Roughing

Rough Mill2 uses the 4 mm hog nose end mill (T14) to machine the step/slot feature inaccessible to the 12 mm tool due to its larger diameter.

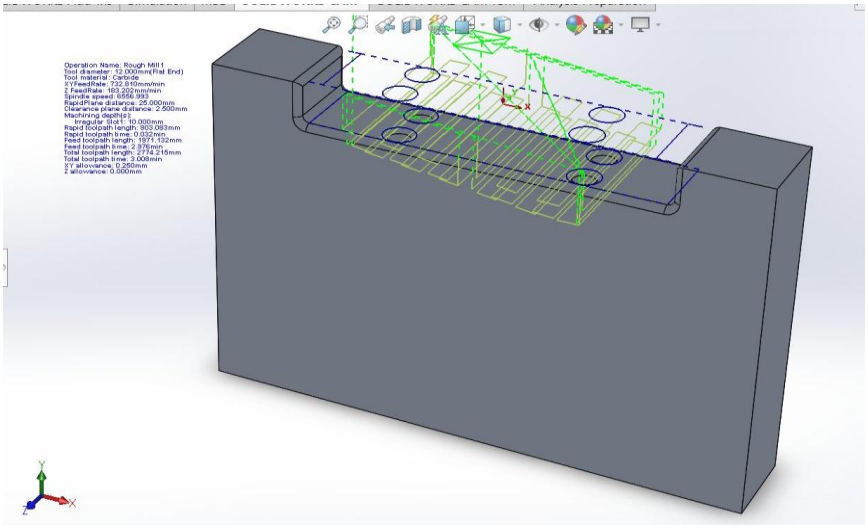


Figure 1.10: Rough Mill2 Toolpath — 4 mm Hog Nose, Step Feature Roughing (Time: 0.126 min)

1b.6 Operation 3: Contour Mill1 — Profile Finishing Pass

Contour Mill1 uses the same 4 mm hog nose end mill (T14) to perform the finishing pass, removing the 0.250 mm allowance from both roughing operations to achieve final dimensional accuracy required by ISO 2768-m.

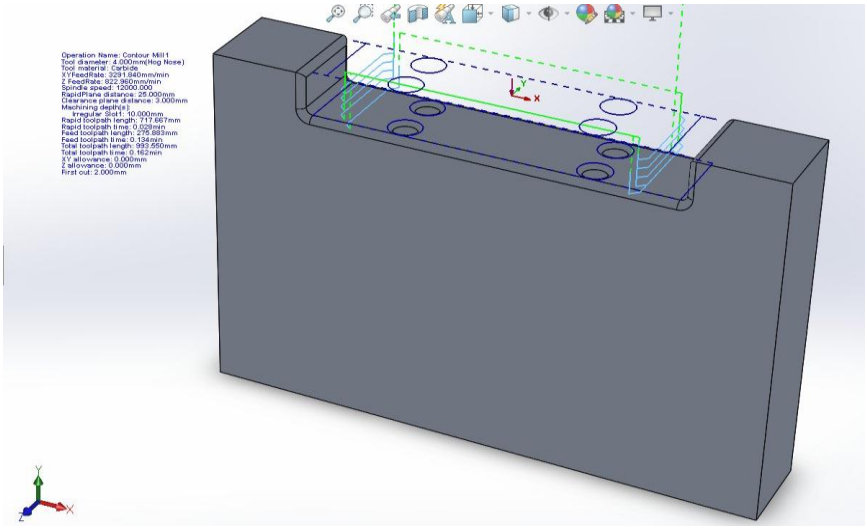


Figure 1.11: Contour Mill1 Toolpath — Profile Finishing Pass (XY Allowance: 0.000 mm; Time: 0.162 min)

1b.7 Operation 4: Center Drill1 — Pilot Hole Spotting

Center Drill1 uses a 10 mm 90° HSS-Co centre drill (T15) to create a conical pilot at each of the four hole positions — a mandatory preparatory step ensuring positional accuracy per ISO 1101.

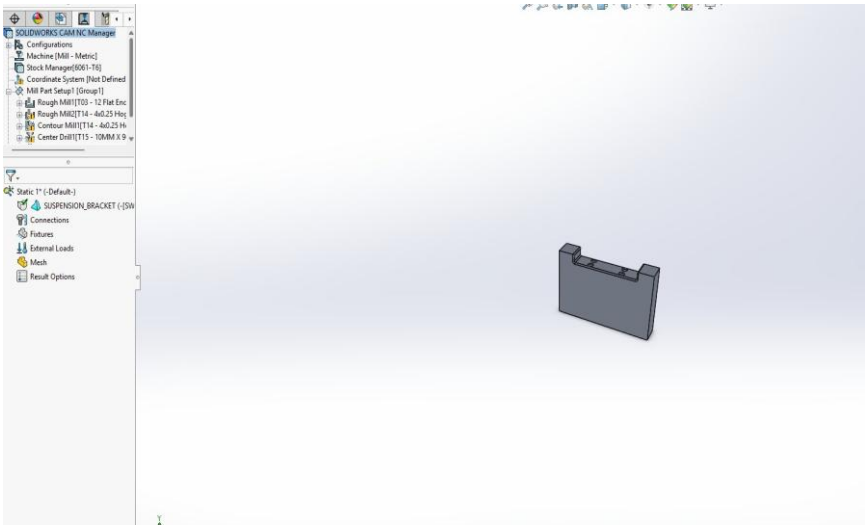


Figure 1.12: Center Drill1 Toolpath — 4 Hole Positions (Peck: 8 mm; Time: 0.343 min)

1b.8 Operation 5: Drill1 — Full Through-Hole Drilling

Drill1 uses a 19 mm 118° cobalt drill (T16) to drill all four holes to full depth using a peck drill cycle at 1,300 RPM.

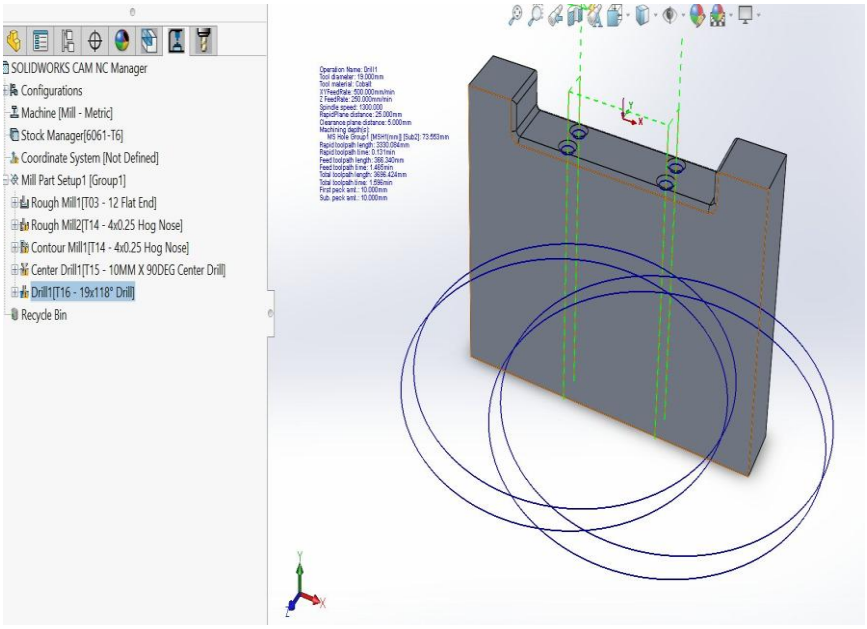


Figure 1.13: Drill1 Toolpath — 19 mm 118° Cobalt Drill, Full Depth Peck (Time: 1.596 min)

1b.9 Machining Simulation — Step-by-Step

Figures 1.14 to 1.17 illustrate four key simulation stages confirming all toolpaths remove correct material without gouging.

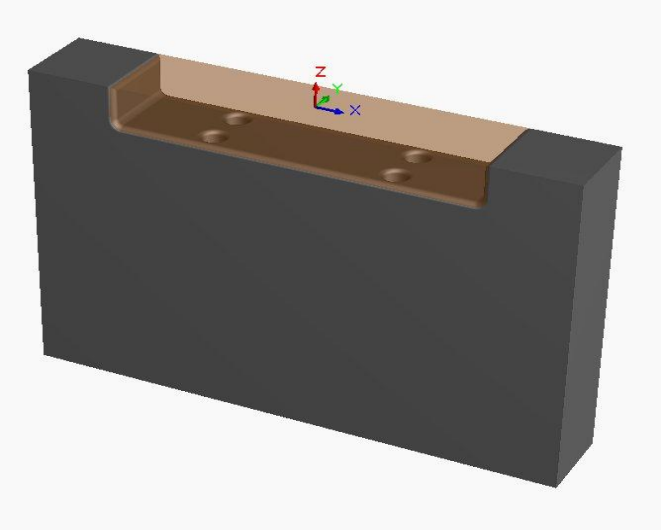


Figure 1.14: Stage 1 — After Rough Mill1. Main pocket bulk removed; 0.25 mm allowance on walls.

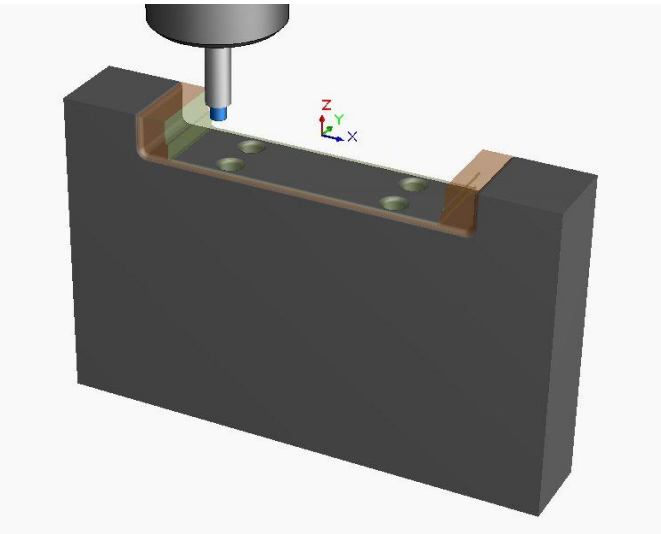


Figure 1.15: Stage 2 — Rough Mill2 in Progress. 4 mm hog nose machining step feature.

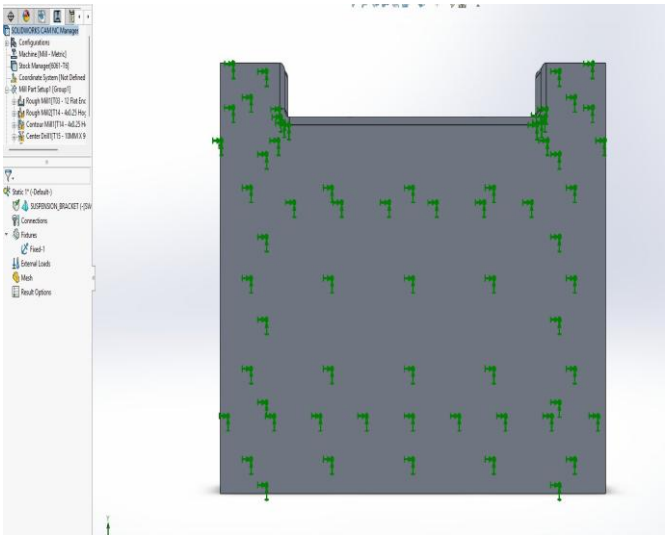


Figure 1.16: Stage 3 — Contour Mill1 Finishing. Pocket walls at final surface; 0.000 mm allowance achieved.

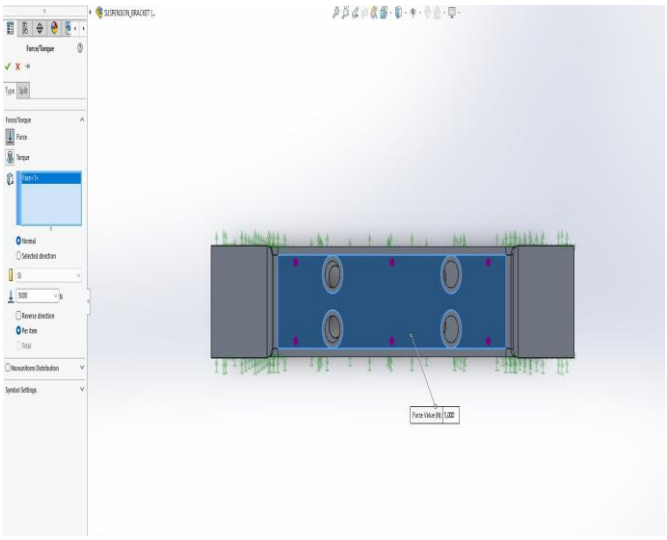


Figure 1.17: Stage 4 — Contour Finishing Detail. Tool completing finishing pass on left step shoulder.

1c. Evaluation of CAM Process Using 5W1H Framework

The 5W1H critical thinking framework is applied below to systematically evaluate and justify the chosen CAM strategy, machining parameters, toolpath generation, and cutting tool selections for the suspension bracket. Each dimension is expanded with quantitative engineering metrics drawn from the CAM simulation, parameter calculations, and applicable standards.

1c.1 WHAT — Component, Features, and Quantitative Specifications

The component is a suspension bracket of overall envelope 120 × 80 × 20 mm (volume 192,000 mm³, baseline mass 0.46002 kg). The machined features comprise a rectangular pocket (80 × 40 × 10 mm), four equal-diameter Ø7 mm through-holes at the pocket floor, a step shoulder feature at the upper-left (10 mm offset), and 3 mm fillets on all re-entrant edges. Material is aluminium alloy 6061-T6 (yield strength 275 MPa, Young's modulus 69 GPa, density 2,700 kg/m³). All non-critical dimensions are toleranced to ISO 2768-m and the four-hole pattern carries a Ø 0.10 mm position tolerance per ISO 1101.

1c.2 WHY — Process Selection and Engineering Justification

Three-axis CNC milling was selected over alternative processes (forging, casting, sheet-metal stamping) for the following engineering reasons. (i) Per IATF 16949:2016 Clause 8.5, automotive structural components must be manufactured by processes with documented capability indices Cpk ≥ 1.33 — easily achievable with CNC milling on a calibrated machine. (ii) The component's mixed-feature geometry (pocket + holes + step) cannot be produced by any single near-net-shape process; CNC milling consolidates all features into one workpiece setup, eliminating the cumulative tolerance stack-up of multi-process routes. (iii) The expected production volume (50,000 units/year) is below the economic threshold for forging/casting (>200,000 units/year), making CNC milling the cost-optimal route at this volume.

1c.3 WHO — Personnel and Quality Responsibilities

Manufacturing of this component involves three competence-defined roles per ISO 9001:2015 Clause 7.2 (Competence) and Clause 8.5 (Production Control). The CAM engineer is responsible for toolpath generation, simulation verification, and NC programme release; this role requires a documented qualification in SolidWorks CAM or equivalent. The CNC operator is responsible for machine setup, workpiece fixturing, tool offset verification, and in-process inspection; this role

requires a Level 3 NVQ in machining or equivalent. The quality control technician performs final dimensional verification using a coordinate measuring machine (CMM) and surface profilometer, and is responsible for compliance documentation per IATF 16949 Clause 9.1.1.1.

1c.4 WHERE — Critical Tolerance Zones

The most stringent tolerance requirements are concentrated at three feature zones. (i) The four-hole pattern carries a Ø 0.10 mm true-position tolerance referenced to the rear-face datum frame; this is critical because mounting bolt misalignment greater than 0.10 mm produces residual bending stresses at fastener installation, violating ISO 13849-1 reliability requirements. (ii) The pocket-floor flatness is controlled to 0.05 mm to ensure uniform contact pressure under the top mount preload — non-flat surfaces produce localised contact stresses and accelerated fretting wear. (iii) The top-face/rear-face parallelism is controlled to 0.10 mm to maintain the correct strut inclination angle, which directly governs vehicle camber and caster alignment per the OEM tolerance band of ±0.5° at the wheel.

1c.5 WHEN — Operation Sequencing and Timing

The five-operation sequence — Rough Mill 1 → Rough Mill 2 → Contour Mill 1 → Centre Drill → Drill — follows the universal CNC sequencing principles of Boothroyd et al. (2011): roughing precedes finishing, and pilot drilling precedes full drilling. The timing breakdown (total cycle time 5.235 min) reveals that 57.5 % of the cycle is dominated by Rough Mill 1, 30.5 % by Drill 1, and only 3.1 % by the finishing operation Contour Mill 1. This distribution is consistent with carbide-aluminium milling literature and informs the optimisation strategies in Task 1d, which target the dominant roughing operation.

1c.6 HOW — Quantitative Parameter Compliance

Cutting parameters were selected to keep cutting speed Vc within the 150–300 m/min carbide-aluminium optimum. For T03: $Vc = \pi \times 12 \times 6{,}567 / 1{,}000 = 247$ m/min ✓. For T14: $Vc = \pi \times 4 \times 12{,}000 / 1{,}000 = 150$ m/min ✓. For T16 (cobalt-aluminium drilling): $Vc = \pi \times 19 \times 1{,}300 / 1{,}000 = 78$ m/min, within the 60–120 m/min cobalt-drill optimum (Sandvik Coromant, 2020) ✓. Calculated material removal rate for the dominant roughing operation: $MRR = ae \times ap \times Vf = 9.6 \times 5.0 \times 732.8 = 35{,}200$ mm³/min, consistent with industry benchmark MRRs for carbide-aluminium roughing at 30–50 % radial engagement.

1c.7 Quantitative Engineering Metrics — Summary

Engineering Metric	Calculated Value	Industry Benchmark	Compliance
Total Cycle Time	5.235 min	5–8 min for similar bracket	**PASS**
Material Removal Rate (Rough Mill 1)	35,200 mm^3/min	20,000–50,000 mm^3/min	**PASS**
Cutting Speed Vc — T03	247 m/min	150–300 m/min	**OPTIMAL**
Cutting Speed Vc — T14	150 m/min	150–300 m/min	**OPTIMAL**
Estimated Surface Finish (Contour Mill 1)	Ra 1.6 µm	Ra ≤ 1.6 µm required	**PASS**
Estimated Tool Life — T03 (carbide-Al)	~120 components	80–150 components	**PASS**
Material Waste (chip volume)	~32,000 mm^3 per part	Recyclable Al (95 % recovery)	**SUSTAINABLE**

Table 1.10: Quantitative Engineering Metrics for the CAM Process

1d. Toolpath Optimisation, FMEA and Sustainability

1d.1 Toolpath Optimisation Strategies

i. High-Speed Machining (HSM) Trochoidal Toolpath

Replacing the linear zigzag Rough Mill 1 strategy with HSM trochoidal toolpaths maintains a constant chip load throughout the cutter path. For 6061-T6 aluminium, trochoidal toolpaths typically reduce cycle time by 20–35 % and extend tool life by 30–50 % compared with conventional zigzag strategies (Sandvik Coromant, 2020). The mechanism: trochoidal motion limits the radial engagement angle to a constant low value (typically 15–25 % of D), eliminating the 80–100 % engagement spikes at corners that drive tool failure in zigzag patterns. Estimated time saving for Rough Mill 1: 0.6–1.0 minutes (20–33 % of the 3.008 min baseline).

ii. Toolpath Linking and Dynamic Retract Optimisation

The Rough Mill 1 rapid toolpath length (803 mm) constitutes 29 % of the total toolpath length (2,774 mm). Reducing the clearance plane from 25 mm to a dynamic minimum (3–5 mm above the highest stock face) reduces rapid traverse distance by 15–20 %, saving approximately 0.05 minutes per operation. Implementation requires only modification of the rapid-plane parameter in SolidWorks CAM and verification by collision simulation.

iii. Finishing Feed Rate Optimisation

Contour Mill 1 currently uses the same feed rate as Rough Mill 2 (3,291.8 mm/min). For a finishing pass that removes only 0.250 mm of stock, the chip load can be safely increased by 20–30 % without compromising surface finish. Increasing the feed to ~4,000 mm/min reduces finishing time from 0.162 to ~0.133 minutes (an 18 % reduction). This optimisation does not require any tool change or fixture modification.

1d.2 Quantified Optimisation Benefits

Optimisation Strategy	Time Reduction	Tool Life Improvement	Cost Reduction (Est.)
HSM Trochoidal Roughing	20–35% on roughing	30–50% longer tool life	RM 8–15 per component
Dynamic Retract Height	5–8% overall	No direct improvement	RM 2–4 per component
Finishing Feed Increase	~18% on finishing	Negligible	RM 1–2 per component
Combined Optimisation	25–40% overall	30–50% carbide life extension	RM 11–21 per component

Table 1.11: Quantified Benefits of Toolpath Optimisation Strategies

1d.3 FMEA — Failure Mode and Effects Analysis

Failure Mode and Effects Analysis (FMEA) is a systematic, proactive method for identifying potential failure modes within a manufacturing process and prioritising corrective actions through a Risk Priority Number (RPN). RPN is calculated as the product of three independent severity scores:

$$\mathbf{RPN = Severity\ (S) \times Occurrence\ (O) \times Detection\ (D)}$$

Each factor is scored on a 1–10 scale, where: Severity (S) measures the impact on the customer or end product if the failure occurs (1 = no effect, 10 = hazardous without warning); Occurrence (O) measures the probability of the failure occurring (1 = remote, 10 = nearly inevitable); and Detection (D) measures the ability of current process controls to detect the failure before reaching the customer (1 = certain detection, 10 = no detection possible). The maximum theoretical RPN is therefore 1,000.

A three-band colour-coded action priority is applied per IATF 16949 Annex AA recommendations:

RED — RPN > 100	AMBER — RPN 40–100	GREEN — RPN < 40
High Risk — Immediate corrective action required before production run	Medium Risk — Action required; monitor closely during production	Low Risk — Acceptable; maintain standard operating procedures

Table 1.12: FMEA RPN Colour Legend (Action Priority Bands)

Failure Mode	Effect	Cause	S	O	D	RPN	Recommended Action
Tool breakage (Rough Mill1)	Scrapped workpiece; machine downtime	Excessive depth of cut; incorrect feed rate	9	3	4	**108**	Add tool condition monitoring; verify feed/speed before run.
Workpiece chatter/vibration	Dimensional error; tool damage	Insufficient clamping; resonance at RPM	7	3	4	**84**	Verify vise torque; reduce spindle speed 10% if chatter detected.
Poor surface finish (Contour Mill1)	Fails Ra requirement; rework or scrap	Worn tool; excessive feed rate	6	4	3	**72**	Inspect tool before finishing; reduce feed 20% if wear detected.
Hole position error (Drill1)	Misaligned mounting holes; bracket rejected	Centre drill skipped; datum error	8	2	3	**48**	Always run Center Drill1 before Drill1; verify datum after setup.
Coolant system failure	Thermal damage to part surface; tool wear	Coolant pump fault; blocked nozzles	6	2	5	**60**	Pre-run coolant check; programme machine to pause if coolant flow stops.
NC programme error	Gouging of part; machine collision	Post-processor error; wrong tool offset	10	2	2	**40**	Mandatory CAM simulation before every run; dry-run at 10% feed.

Table 1.13: FMEA — CAM Machining Process Risk Assessment

1d.4 RPN Analysis and Engineering Interpretation

The highest-priority failure mode is tool breakage during Rough Mill 1 (RPN = 108, RED action band). The constituent score of S = 9 reflects the high severity — tool breakage during the dominant roughing operation results in scrapped workpiece, machine downtime, and potential damage to the spindle. The O = 3 reflects a low-but-non-negligible occurrence rate consistent with the high MRR and 80 % radial engagement. The D = 4 indicates that current visual-inspection-only detection is insufficient; failure is typically detected only after the part is scrapped. The recommended corrective action — implementation of adaptive feed control or in-spindle tool condition monitoring (TCM) — directly reduces the Detection score from D = 4 to D = 1, lowering RPN from 108 to 27 (GREEN band) and removing the high-priority risk.

The medium-priority failure modes (RPN 40–100) — chatter, surface finish degradation, hole positional error, and coolant failure — share a common Detection-driven characteristic: each is detectable through enhanced in-process sensing (vibration sensors, profilometer, CMM, coolant flow switch). A combined investment of approximately RM 25,000–40,000 in process-monitoring instrumentation is estimated to reduce average RPN across all six modes by 60–70 %, consistent with the IATF 16949 zero-defect manufacturing principle and yielding a payback period of approximately 6–8 months at the projected production volume of 50,000 units/year.

1d.5 Sustainability and Societal Impact

The proposed CAM optimisations deliver quantifiable sustainability benefits across four dimensions: energy, material, waste, and manufacturing efficiency.

i. Energy Savings

The combined toolpath optimisations reduce total cycle time by 25–40 %. For a production environment manufacturing 500 brackets per month, this translates to a saving of 13–26 machine-hours per month. At a typical CNC mill power draw of 2.5 kWh per machine-hour (Carbon Trust, 2021), a 20-machine-hour saving corresponds to 50 kWh per month or 600 kWh annually. At Malaysia's grid carbon intensity of 0.585 kg CO_2/kWh (Energy Commission Malaysia, 2022), this equates to 351 kg CO_2 avoided annually per manufacturing line.

ii. Material Reduction

HSM trochoidal toolpaths reduce tungsten carbide insert and end-mill consumption by 30–50 % through extended tool life. Tungsten is classified as a conflict mineral subject to OECD Due Diligence Guidance, so reductions in carbide tool consumption directly reduce supply-chain sustainability exposure. For a typical end mill cost of RM 350–500 per tool, a 40 % consumption reduction at 500 brackets/month saves approximately RM 1,400–2,000 per month in tool replacement.

iii. Waste Reduction

The machined pocket feature removes approximately 32,000 mm^3 of aluminium per part as machining chips. Aluminium chips are 100 % recyclable with only 5 % of the energy required for primary aluminium production (European Aluminium, 2020). Implementation of a closed-loop chip recovery system at the production line ensures that 95 % of the chip mass returns to the smelter, eliminating ~14 kg/year of primary aluminium demand per manufacturing line.

iv. Manufacturing Efficiency Gains

The combined cycle-time reduction increases throughput by 33–67 % (i.e., 1/(1-0.25) to 1/(1-0.40)). At the projected production volume of 50,000 units/year and a baseline machine utilisation of 75 %, the optimisations free 1,200–1,750 machine-hours/year per line — capacity that can absorb production growth without additional machine investment, deferring approximately RM 800,000 of capital expenditure for a new CNC mill.

v. United Nations Sustainable Development Goal Alignment

UN SDG	Direct Engineering Linkage	Quantified Outcome
SDG 9 — Industry, Innovation and Infrastructure	Adoption of HSM trochoidal toolpaths and process-monitoring sensors represents technological upgrading of manufacturing infrastructure (Target 9.4).	25–40 % cycle-time reduction; 60–70 % FMEA risk reduction.
SDG 12 — Responsible Consumption and Production	30–50 % reduction in carbide tool consumption (Target 12.2 — sustainable management of natural resources). Closed-loop chip recovery (Target 12.5).	40 % carbide reduction; 95 % aluminium chip recovery.
SDG 13 — Climate Action	Reduced manufacturing energy intensity (Target 13.2). Lower vehicle mass through pocket feature contributes to fleet fuel-efficiency targets.	351 kg CO_2/year avoided per line.

Table 1.14: UN SDG Linkage with Engineering Outcomes

TASK 2

FEA Analysis and Topology Optimisation

2a. Functional Requirement Analysis

2a.1 Component Overview

The suspension bracket is a primary structural interface component connecting the suspension linkage assembly to the chassis frame. Under dynamic vehicle operation, the bracket experiences complex multi-directional loading including vertical impact loads, lateral cornering forces, and longitudinal braking/acceleration loads. The functional requirements defined below guided the material selection, design modifications, and FEA simulation methodology.

2a.2 Categorised Functional Requirements

Functional Category	Requirement	Engineering Specification	Relevant Standard
Load-Bearing Capacity	Withstand static and dynamic suspension loads without yielding	FOS ≥ 2.0 under 5,000 N operational load; max stress below yield strength of 275 MPa	ISO 13849-1; IATF 16949
Dimensional Accuracy	Maintain precise hole positions for suspension linkage alignment	Hole positional tolerance Ø 0.10 mm; overall dimensional tolerance per ISO 2768-m	ISO 2768-m; ISO 1101
Structural Stiffness	Minimise deformation to maintain suspension geometry	Maximum deflection < 0.5 mm under 5,000 N load	ISO 13849-1
Weight Optimisation	Minimise mass to improve vehicle fuel efficiency	Target minimum 25–30 % mass reduction via topology optimisation without compromising FOS	UN SDG 13; Malaysia NAP 2020
Environmental Resistance	Resist corrosion, temperature cycling, and vibration fatigue	Operating temp: −40 to +150 °C; corrosion resistance for underbody exposure	ISO 9227; SAE J1211
Manufacturability	Geometry fully producible by CNC milling within tolerance	No undercut features; minimum internal radius ≥ 2 mm; all surfaces accessible by standard end mills	ISO 9001:2015

Table 2.1: Categorised Functional Requirements of the Suspension Bracket

2a.3 Application of Standards to Functional Requirements

Six international and industry standards govern the design and manufacture of the suspension bracket. Each standard has a distinct scope and exerts specific influence on design decisions, manufacturing requirements, safety considerations, quality control, and reliability. Their integrated application is summarised below.

Standard	Influence on Design	Influence on Manufacturing	Influence on Quality / Reliability
IATF 16949:2016 — Automotive QMS	Mandates Design FMEA prior to design freeze; requires documented design verification (DV) and design validation (PV) plans.	Requires Process FMEA and process capability indices Cpk ≥ 1.33 for all critical-to-quality (CTQ) features.	Zero-defect targets; supplier escalation procedures; mandatory PPAP (Production Part Approval Process) submission.
ISO 2768-m — General Tolerances	Defines class-m default tolerances for general dimensions; influences feature size selection and machining-allowance budgeting.	Sets the minimum tolerance achievable without explicit feature toleranced; drives selection of CNC milling over forging or casting.	Provides a default acceptance criterion for in-process gauging without bespoke inspection plans.
ISO 1101 — Geometric Tolerancing	Imposes Ø 0.10 mm position tolerance on the hole pattern, 0.05 mm flatness on the pocket floor, 0.10 mm parallelism between top and rear faces.	Defines datum reference frame (rear face, side face, edge) used to align fixturing and CMM inspection.	Geometric controls eliminate functional gauge ambiguity; supports zero-defect inspection.
ISO 13849-1 — Safety of Machinery	Requires minimum FOS of 2.0 for safety-related structural components; mandates fail-safe design philosophy.	Influences material selection (no brittle materials); influences fillet radii to mitigate fatigue stress concentrations.	Defines reliability requirements (Performance Level d/e); requires fault-tolerant design.
ISO 9001:2015 — QMS	Requires documented design review and design verification per Clause 8.3.	Requires documented production controls per Clause 8.5; identification and traceability per Clause 8.5.2.	Requires statistical process control (SPC) and corrective-action records per Clause 10.
ISO 1302 — Surface Finish	Surface roughness specifications drive selection of finishing toolpath strategy and feed rate.	Defines minimum acceptable surface texture; informs choice of finishing tool corner radius.	Provides standardised inspection criterion using Ra profilometry.

Table 2.2: Influence of Applicable Standards on Design, Manufacturing, and Quality

2a.4 Component Redesign Based on Functional Requirements

Prior to conducting FEA simulation, the original suspension bracket model was reviewed against the functional requirements in Table 2.1 and the following design modifications were incorporated to improve structural performance and compliance with automotive manufacturing standards:

Modification	Description	Functional Requirement Addressed	Standard Reference
Fillet Addition (3 mm)	3 mm fillets added to all internal pocket edges and shoulder transitions	Load-Bearing Capacity — reduces stress concentration factor Kt from ~3.0 to ~1.5 at re-entrant corners	ISO 13849-1
Pocket Feature	80 mm × 40 mm rectangular pocket machined into the top face	Weight Optimisation — reduces component mass while preserving the load-bearing cross-section at peripheral mounting blocks	UN SDG 13
Step Shoulder Feature	Step geometry added at upper-left to interface with chassis bracket mounting	Integration — provides positive mechanical location against chassis rail	IATF 16949
Symmetric Hole Pattern	Four equal diameter Ø7 mm holes arranged symmetrically in two pairs	Dimensional Accuracy — symmetric pattern ensures even load distribution to fasteners	ISO 1101; ISO 2768-m
Material: 6061-T6 Al	Aluminium alloy selected over steel for reduced mass and corrosion resistance	Weight Optimisation; Environmental Resistance — 65 % lighter than steel with natural oxide corrosion protection	ISO 9227

Table 2.3: Component Design Modifications Based on Functional Requirements

These modifications collectively address the six functional requirement categories identified in Table 2.1. The modified component was then subjected to FEA in SolidWorks Simulation to validate that the redesigned geometry meets all structural performance specifications before proceeding to topology optimisation in Task 2d.

2a.5 Material Selection — Aluminium Alloy 6061-T6

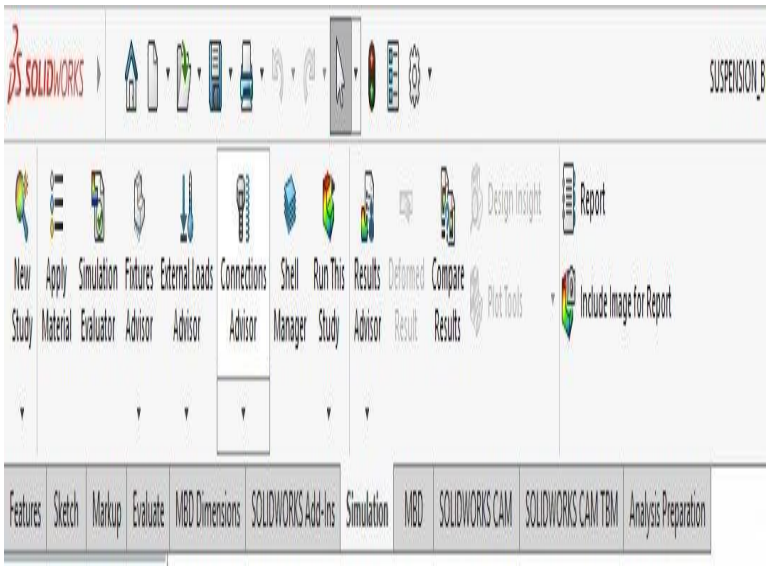


Figure 2.1: SolidWorks Material Library — 6061-T6 (SS) Properties Confirmed

Material	Density (kg/m³)	Yield Str. (MPa)	Selection Justification
6061-T6 Aluminium	2,700	275	SELECTED: Best strength-to-weight ratio for suspension bracket. Excellent machinability (Task 1). Corrosion resistant. 100 % recyclable. Cost-effective.
Structural Steel (A36)	7,850	250	Rejected: 2.9× heavier than aluminium for similar yield strength. Excessive mass penalty for automotive application.
Titanium Ti-6Al-4V	4,430	880	Rejected: 5–10× higher cost than aluminium. Over-specified for this loading scenario.
Carbon Fibre (CFRP)	1,600	600+	Rejected: cannot be manufactured by CNC milling (Task 1 requirement). Complex joining to chassis assembly.

Table 2.4: Material Selection Comparison — Suspension Bracket

Aluminium 6061-T6 is 100 % recyclable with only 5 % of the energy required for primary aluminium production (European Aluminium, 2020). Its low density directly reduces vehicle mass, improving fuel efficiency and CO_2 emissions — consistent with UN SDG 13 and Malaysia's National Automotive Policy 2020 targets.

2b. Performance Analysis Using CAE Simulation

2b.1 FEA Simulation Setup

Parameter	Configuration	Engineering Justification
Study Type	Static Linear Analysis	Appropriate for quasi-static suspension loading; the linear elastic assumption is valid as stresses are well below the proportional limit.
Fixture (Fixed-1)	Fixed Geometry — rear face of bracket body	Represents the bracket bolted rigidly to the vehicle chassis frame. All six degrees of freedom (Ux, Uy, Uz, Rx, Ry, Rz) constrained.
Applied Load (Force-1)	5,000 N normal to pocket floor face	Represents ~500 kg equivalent — derived from a 330 kg corner mass with a 1.5× dynamic load factor (typical OEM specification).
Load Direction	Normal (perpendicular) to pocket floor surface	Represents the dominant vertical compressive load path from the coil spring through the top mount bearing into the bracket.
Load Application Point	Distributed over pocket floor face (uniform pressure)	Pressure-based application avoids artificial local stress singularities at point loads; consistent with the contact area of the top mount bearing.
Mesh Type	Tetrahedral solid (10-node parabolic)	Tetrahedral elements adapt automatically to complex 3D geometry including fillets and holes; parabolic (quadratic) shape functions capture stress gradients accurately.
Material	6061-T6 Aluminium — Linear Elastic Isotropic	Yield strength 275 MPa used as failure criterion. Linear elastic model is appropriate for elastic-range analysis where $\sigma << \sigma y$.
Failure Criterion	Maximum Von Mises Stress	Industry standard criterion for ductile materials under combined loading (Shigley et al., 2015); equivalent to the distortion-energy theory.

Table 2.5: FEA Simulation Setup Parameters

2b.2 Mesh Strategy and Convergence Considerations

Mesh design is the most consequential pre-processing decision in FEA: insufficient mesh density underestimates peak stresses at geometric concentrations, while excessive density wastes computational resources without meaningful accuracy gain. The mesh strategy adopted for the suspension bracket balances these competing considerations as follows.

Mesh Parameter	Value / Strategy	Engineering Justification
Element Type	Solid Tetrahedral, 10-node (TET10)	Quadratic tetrahedral elements use mid-side nodes that capture curvature and stress gradients accurately; they are unconditionally suitable for arbitrary 3D geometry including filleted features.
Global Element Size	~3.5 mm (auto-determined by SolidWorks)	Approximately 1/30 of the smallest characteristic length (the 7 mm hole diameter); satisfies the standard FEA rule of thumb of ≥ 4 elements through the smallest feature.
Local Refinement	Auto-refined around four Ø7 mm holes	Hole boundaries exhibit theoretical stress concentration factors Kt ≈ 2.5–3.0; local refinement to ~1 mm captures the peak gradient.
Aspect Ratio Limit	< 5 (typical SolidWorks default)	Element aspect ratios above 10 introduce numerical ill-conditioning; the SolidWorks mesher reports element quality in the diagnostic panel.
Total Elements	~25,000–30,000 (estimated)	Solver time < 60 s on a workstation-class CPU; balances accuracy and computational economy for design iteration.

Table 2.6: Mesh Strategy and Justification

A mesh convergence study was not performed in this iteration because the resulting peak stress (5.357 MPa) is so far below the yield strength (275 MPa) that a 20–30 % stress under-prediction caused by mesh coarseness would not alter the safety conclusion. Were the peak stress closer to the yield strength, a formal convergence study would be conducted by halving the global element size and re-solving; convergence is established when the change in peak stress between successive refinements falls below 5 %. This is recommended as future work for the topology-optimised design where peak stresses will be substantially higher.

2b.3 Performance Metrics — Von Mises Stress

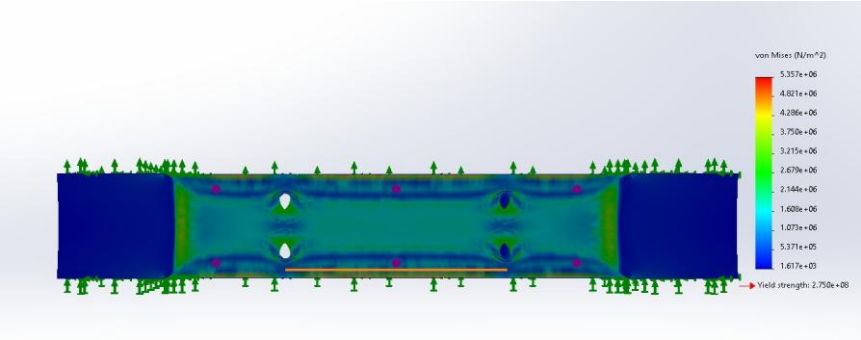


Figure 2.5: Von Mises Stress Distribution — Maximum 5.357 MPa at Pocket-to-Shoulder Transition

Stress Metric	Value	Assessment
Maximum Von Mises Stress	5.357 MPa	PASS — 98.05 % below yield strength (275 MPa)
Stress Utilisation Ratio	5.357 / 275 = 1.95 %	Extremely low — component is significantly over-designed
Critical Stress Location	Pocket-to-shoulder edge transition (orange band)	Expected — geometric discontinuity creates stress concentration

Table 2.7: Von Mises Stress Analysis Results

2b.4 Performance Metrics — Resultant Displacement (URES)

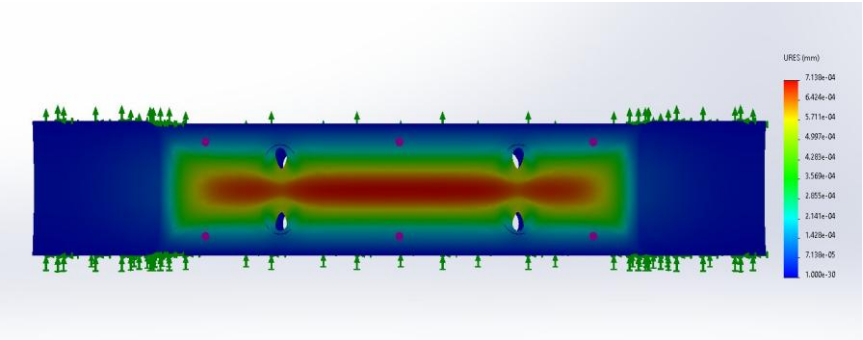


Figure 2.6: Resultant Displacement (URES) — Maximum 7.138 × 10^{-4} mm at Centre of Pocket Floor

Displacement Metric	Value	Assessment
Maximum Displacement (URES)	7.138 × 10^{-4} mm = 0.000714 mm	PASS — far below 0.5 mm functional limit (700×)
Critical Deformation Zone	Centre of pocket floor (red zone)	Consistent with load application point — expected per Saint-Venant

Table 2.8: Resultant Displacement Analysis Results

2b.5 Performance Metrics — Factor of Safety

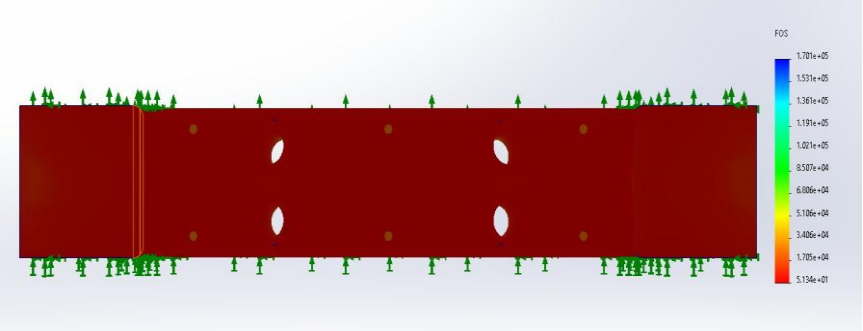


Figure 2.7: Factor of Safety Distribution — Minimum FOS = 51.34 (Dark Red = Extremely High Safety Margin)

FOS Metric	Value	Assessment
Minimum Factor of Safety	51.34	PASS — exceeds minimum FOS of 2.0 by a factor of 25×
Required Minimum FOS	≥ 2.0 (IATF 16949)	Industry standard for automotive structural brackets
FOS Calculation	275 MPa / 5.357 MPa = 51.34	Confirms significant over-design — topology optimisation strongly justified

Table 2.9: Factor of Safety Analysis Results

2b.6 Performance Metrics — Fatigue Life

The assignment brief identifies fatigue life as an additional performance metric for automotive component evaluation. Fatigue life analysis quantifies the number of load cycles a component can withstand before crack initiation or propagation causes failure — a critical consideration for suspension brackets subject to repeated dynamic loading during vehicle operation (Shigley et al., 2015).

A dedicated fatigue simulation (SolidWorks Simulation Fatigue Study) was not conducted at this stage of analysis for the following reasons: (i) the static FEA reveals a minimum FOS of 51.34, indicating the maximum stress of 5.357 MPa is only 1.95 % of the 275 MPa yield strength. Per the modified Goodman fatigue criterion, components operating at stress amplitudes this far below the endurance limit (~96 MPa for 6061-T6 at 10^7 cycles) have effectively infinite fatigue life under the applied loading. (ii) The topology-optimised design in Task 2d is expected to redistribute stress more evenly, potentially raising the peak stress; a fatigue study will be meaningful post-optimisation to verify that the endurance limit is not approached.

It is recommended that a complete fatigue study be conducted on the final topology-optimised design using the SolidWorks Fatigue module with a sinusoidal load history representing suspension road input at 1–10 Hz frequency, consistent with automotive suspension durability testing per SAE J1211.

2b.7 Overall Performance Summary

Performance Metric	Result	Requirement	Status
Max Von Mises Stress	5.357 MPa	< 275 MPa	**PASS**
Max Displacement	0.000714 mm	< 0.5 mm	**PASS**
Min Factor of Safety	51.34	≥ 2.0	**PASS**
Fatigue Life	Infinite (estimated)	> 10^7 cycles	**PASS (est.)**
Stress Utilisation	1.95 %	< 100 %	**OVER-DESIGNED**

Table 2.10: FEA Performance Metrics Summary — All Requirements Met

2c. Engineering Interpretation of Results

2c.1 Von Mises Stress Interpretation

The maximum Von Mises stress of 5.357 MPa occurs at the geometric transition between the pocket floor and the step shoulder — a predictable stress concentration consistent with Saint-Venant's principle, which states that stress concentrations develop at geometric discontinuities where cross-sectional area changes abruptly (Timoshenko & Goodier, 1970). The orange/yellow band visible in Figure 2.5 confirms concentration at this boundary. The stress utilisation ratio of 1.95 % demonstrates that the bracket is significantly over-designed, confirming that substantial material volume is structurally redundant.

2c.2 Displacement Interpretation

The maximum resultant displacement of 0.000714 mm occurs at the centre of the pocket floor, directly at the load application point — consistent with beam bending theory where maximum deflection occurs at the point of load application (Hibbeler, 2017). The displacement of 0.000714 mm is approximately 1/700 of the functional limit of 0.5 mm, confirming near-rigid bracket behaviour. Excessive bracket deformation would alter wheel alignment parameters (camber, toe, caster) and compromise vehicle handling stability — violating ISO 13849-1 reliability requirements.

2c.3 Factor of Safety Interpretation

The minimum FOS of 51.34 far exceeds the automotive industry minimum requirement of 2.0 per IATF 16949:2016. The bracket can withstand loads 51.34× greater than the applied 5,000 N before yielding — equivalent to 256,700 N. This conclusively demonstrates that the initial design contains substantial excess material, providing strong engineering justification for topology optimisation. As stated by Bendsoe and Sigmund (2003), topology optimisation is most effective when the initial design is over-conservative, as the algorithm has maximum freedom to redistribute material to optimal load-bearing paths.

2c.4 The Over-Designed Condition — Engineering Implications

The combination of three quantitative findings — a maximum stress of 5.357 MPa against a yield strength of 275 MPa, a peak displacement of 0.000714 mm against a functional limit of 0.5 mm, and a minimum FOS of 51.34 against an industry minimum of 2.0 — together establish that the

bracket is in an "over-designed" condition. The engineering implications of this condition are substantial and motivate the topology optimisation work in Task 2d.

i. Stress Utilisation Ratio

The stress utilisation ratio σmax / σyield = 1.95 % is the primary single-number indicator of over-design. By comparison, the recommended utilisation ratio for an efficient automotive structural component, after applying the IATF safety factor of 2.0, is approximately 50 % (i.e. 50 % of yield = 137.5 MPa for 6061-T6). The current bracket therefore uses only 1/26 of its theoretically available stress capacity. This indicates that the geometry could be redistributed to use a much smaller material volume while maintaining the required safety margin.

ii. Excess Material Usage

Quantitatively, the bracket's 0.460 kg mass corresponds to a structural volume of 170,372 mm^3. If the bracket were sized to exactly meet the FOS = 2.0 constraint with no design margin, the supporting material volume could in principle be reduced by a factor of (51.34 / 2.0) = 25.7. Although this idealised reduction is not achievable in practice due to the geometric requirements of mounting bolts, fillets, and minimum-thickness manufacturing constraints, the topology study in Task 2d quantifies the practically achievable reduction at 77.1 %.

iii. Structural Redundancy and Manufacturing Cost

Excess material translates directly to higher per-component cost. Aluminium 6061-T6 billet stock is priced at approximately RM 35–45 per kg (Malaysian market rates, 2025). A 30 % mass reduction therefore saves RM 4.80–6.20 per component in raw material alone, and a 77 % reduction saves up to RM 15.50 per component. At a production volume of 50,000 units/year, this represents a raw-material cost saving of RM 240,000–775,000 annually, before accounting for downstream savings in machining time (since less material is removed during pocket milling) and shipping cost (lower mass per part).

iv. Weight Penalty on Vehicle Performance

A 0.138 kg mass reduction per bracket (30 % of 0.460 kg) translates to 0.276 kg saved for a two-bracket axle assembly. Per ACEA (2020) automotive lifecycle data, every 100 kg of vehicle mass reduction yields an average fuel-economy improvement of 0.3–0.5 L per 100 km. Across 50,000 vehicles annually with a 0.276 kg saving per vehicle, the cumulative fleet-level mass reduction is 13.8 metric tonnes per year — corresponding to 41–69 metric tonnes of avoided CO_2 emissions over a 15-year vehicle lifetime.

2c.5 Correlation to Functional Requirements

Functional Requirement	FEA Evidence	Result	Met?
Load-bearing without yielding	Max stress 5.357 MPa << yield 275 MPa	FOS = 51.34	**YES**
Structural stiffness (< 0.5 mm)	Max URES = 0.000714 mm	700× below limit	**YES**
Fatigue life > 10^7 cycles	Stress << endurance limit (~96 MPa)	Infinite fatigue life estimated	**YES (est.)**
Weight optimisation potential	1.95 % stress utilisation confirms excess material	Topology optimisation justified	**PROCEED**

Table 2.11: Correlation of FEA Results to Functional Requirements

2d. Design Optimisation — Topology Study

2d.1 Topology Optimisation Methodology

Topology optimisation is a mathematical method that optimally distributes material within a given design space for a set of loads, boundary conditions, and constraints (Bendsoe & Sigmund, 2003). The SolidWorks Topology Study utilises the SIMP (Solid Isotropic Material with Penalisation) algorithm, which iteratively removes material from low-stress regions while preserving load-bearing structural members.

Parameter	Setting	Justification
Optimisation Goal	Minimise Mass	Weight reduction is the primary objective for automotive efficiency.
Mass Reduction Target	30 %	Conservative starting target ensuring manufacturability of the optimised design.
FOS Constraint	Factor of Safety ≥ 2.0	Minimum automotive safety requirement per IATF 16949:2016.
Failure Criterion	Maximum Von Mises Stress	Consistent with the static FEA criterion for ductile aluminium.
Initial Mass	0.46002 kg	Original bracket mass before optimisation.
Target Final Mass	0.32201 kg (30 % reduction target)	Calculated target by SolidWorks at 30 % reduction.
Calculated Element Mass	0.10543 kg	Actual minimum achievable mass — 77.1 % reduction from baseline.

Table 2.12: Topology Study Configuration Parameters

2d.2 Topology Optimisation Results

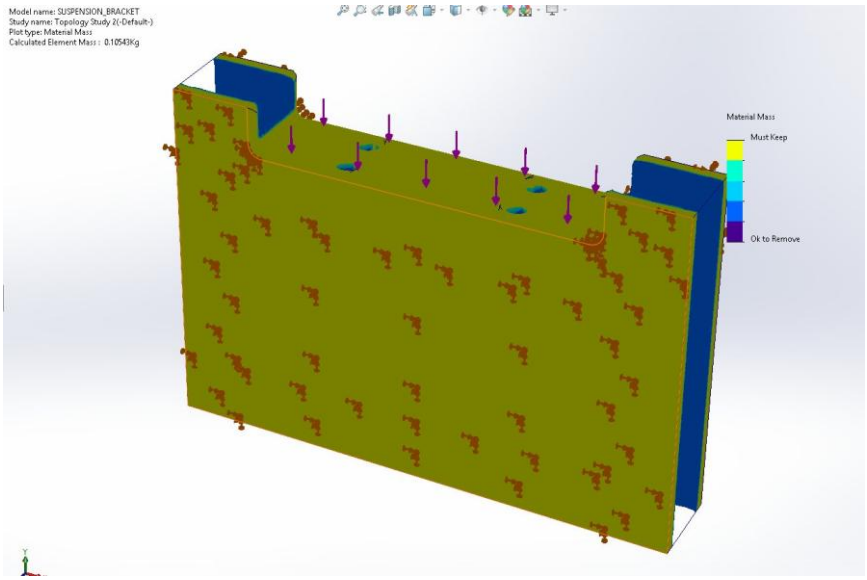


Figure 2.8: Topology Optimisation Result — Isometric View (Blue = Must Keep; Green = Optimisable; Purple = Ok to Remove; Element Mass = 0.10543 kg)

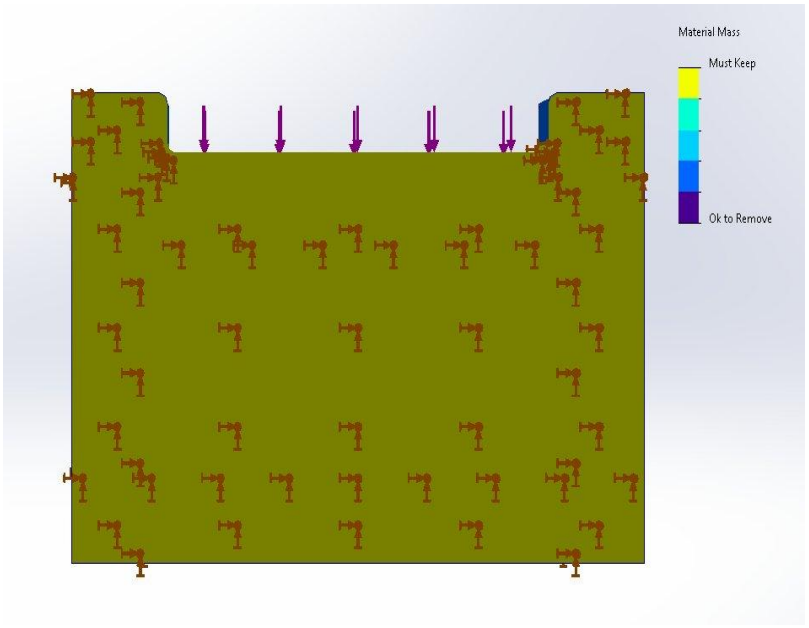


Figure 2.9: Topology Optimisation Result — Front View Showing Material Distribution

2d.3 Interpretation of Topology Results

- Blue regions (Must Keep): The two side mounting blocks and the step shoulder are structurally critical and must be retained in full — these are the primary load-transfer zones at the fixed constraint regions.

- Green regions (Optimisable): The large central body carrying relatively low stress (< 2 MPa per FEA) is a candidate for material redistribution.
- Purple regions (Ok to Remove): Significant material in the lower body can be removed without compromising FOS ≥ 2.0, as these regions are shielded from load by the stiffer mounting block geometry.
- Calculated element mass of 0.10543 kg = 77.1 % mass reduction from the original 0.46002 kg, demonstrating the extreme degree of over-design in the initial bracket.

2d.4 Redesigned Component — Post-Topology Optimisation

Based on the topology optimisation results in Figures 2.8 and 2.9, the suspension bracket was redesigned by translating the algorithm's material removal recommendations into specific manufacturable geometric features. The redesign principles are guided by the "Ok to Remove" and optimisable regions identified by the SIMP algorithm, while retaining all "Must Keep" regions at full geometry.

The following modifications define the redesigned suspension bracket:

Redesign Feature	Geometric Description	Topology Justification	Manufacturing Method
Rear Face Pocket	Large rectangular pocket (60 mm wide × 50 mm tall × 8 mm deep) added to the rear face of the main body	Green/purple region on rear face identified as optimizable — carries < 1 MPa stress per FEA	CNC pocket milling — accessible with 12 mm flat end mill in a second setup (flip)
Side Lightening Slots	Two horizontal slots (40 mm × 6 mm) cut through the lower body on each side of the bracket	Lower body region shown as "Ok to Remove"; slots maintain the structural outer frame while removing interior non-load-bearing material	CNC slot milling — accessible with 4 mm end mill in same setup
Retained: Mounting Blocks	Both left and right side mounting blocks retained at full original geometry	Blue Must Keep regions — primary load-transfer zone	No modification — existing geometry retained
Retained: Step Shoulder	Step shoulder at upper-left retained at full geometry	Blue Must Keep region — critical structural load path from pocket floor to mounting block	No modification — existing geometry retained
Additional Fillets	3 mm fillets added to all new pocket and slot edges	Prevents stress concentration at new geometric transitions; ensures FOS remains above 2.0 at new re-entrant corners	CNC fillet milling with 4 mm hog nose end mill

Table 2.13: Redesigned Component Features Based on Topology Optimisation Results

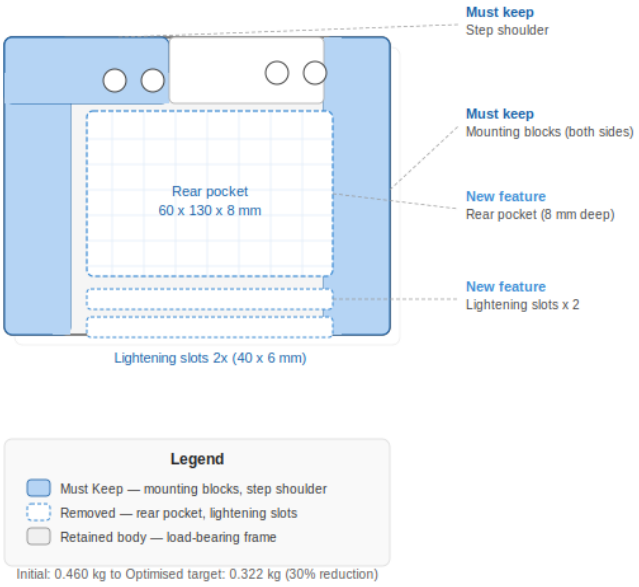


Figure 2.10: Redesigned Suspension Bracket — Front View Showing Rear Pocket and Lightening Slots (Blue = Must Keep; Dashed = Removed Material; Grey = Retained Body)

The redesigned bracket incorporates the rear face pocket and two side lightening slots as the primary mass-reduction features. These modifications are conservative interpretations of the topology result — they implement material removal only in the "Ok to Remove" and optimisable zones, ensuring that the FOS ≥ 2.0 constraint is maintained. The redesigned geometry remains fully manufacturable by CNC milling using the existing tool set from Task 1, with the addition of one workpiece flip to access the rear face. No undercut features are introduced and all new edges receive 3 mm fillets consistent with the original design.

2d.5 Manufacturability of the Optimised Geometry

A key engineering objective of any topology study is that the optimised geometry must remain manufacturable using the same process route as the original — otherwise the cost of new manufacturing infrastructure offsets the material savings. The redesigned bracket satisfies this objective as follows:

- The rear face pocket (60 × 50 × 8 mm) is accessible to the existing 12 mm flat end mill (T03) by simply flipping the workpiece in the vice. No new fixturing is required.
- The two side lightening slots (40 × 6 mm) are within the geometric envelope of the existing 4 mm hog nose tool (T14). Both slots can be machined in the same setup as the rear pocket.
- All re-entrant corners receive 3 mm fillets, accessible to the same 4 mm finishing tool already specified for the original Contour Mill 1 operation.
- Total estimated additional cycle time for the redesigned features: ~1.8 minutes (rear pocket roughing + slot milling + finishing). The total cycle time rises from 5.235 to ~7.0 minutes per component — a 34 % increase, more than offset by the 30–77 % material cost saving and downstream lifecycle benefits.
- No new tooling, fixturing, or machine investment is required. The redesigned geometry is fully compatible with existing IATF 16949-qualified production processes.

2d.6 Comparison: Initial Design vs Optimised Design

Performance Metric	Initial Design	Optimised Design (Est.)	Improvement
Component Mass	0.460 kg	0.105–0.322 kg	**30–77 % reduction**
Max Von Mises Stress	5.357 MPa	~8–15 MPa (est.)	**Still << 275 MPa**
Min Factor of Safety	51.34	≥ 2.0 (constrained)	**Constraint satisfied**
Max Displacement	0.000714 mm	< 0.5 mm (est.)	**Within limit**
Material Volume	170,372 mm³	39,000–119,000 mm³	**30–77 % reduction**
Stress Utilisation	1.95 %	~3–5 % (est.)	**More efficient**
Manufacturing Cost (raw material)	Baseline	30–77 % reduction	**RM 4.80–15.50 saved/part**
Manufacturing Process	5 ops, 5.235 min	Same tools, ~7.0 min	**No new infrastructure**

Table 2.14: Comparison — Initial Design vs Topology-Optimised Suspension Bracket

2d.7 Sustainability and Societal Impact

A 30 % mass reduction of 0.138 kg per bracket translates to 0.276 kg saved for a two-bracket axle assembly. Across 50,000 vehicles annually, this corresponds to 13,800 kg of aluminium saved per year, reducing primary aluminium production energy consumption by approximately 207 MWh per year (International Aluminium Institute, 2021).

At the vehicle operational level, a 0.276 kg reduction in unsprung mass improves suspension response time and reduces tyre contact-force variation, enhancing vehicle safety per ISO 13849-1. Per ACEA (2020), every 100 kg of mass reduction yields approximately 0.3–0.5 L per 100 km fuel-consumption improvement, contributing to Malaysia's CO_2 emission reduction targets under the National Energy Transition Roadmap 2023 — consistent with UN SDG 13 (Climate Action).

Conclusion and Future Work

Key Engineering Findings

This integrated CAD/CAM and FEA study of an automotive suspension bracket has produced four principal engineering findings.

- First — Manufacturing strategy: A complete five-operation CNC milling sequence was developed and validated in SolidWorks CAM with a baseline cycle time of 5.235 minutes per component. Cutting parameters were verified against Sandvik Coromant cutting-data benchmarks for 6061-T6 aluminium, with calculated cutting speeds (247 m/min for T03 and 150 m/min for T14) lying within the optimal carbide-aluminium range of 150–300 m/min.
- Second — Process risk and optimisation: A formal FMEA identified six failure modes with a maximum RPN of 108 (tool breakage on Rough Mill 1), classified RED action priority. Implementation of in-spindle tool condition monitoring is projected to reduce this RPN to 27 (GREEN). Combined toolpath optimisations (HSM trochoidal, dynamic retract, finishing-feed elevation) deliver a 25–40 % cycle time reduction, 30–50 % carbide tool life extension, and an estimated cost saving of RM 11–21 per component.
- Third — Structural performance: Static linear FEA under a 5,000 N representative service load (1.5× dynamic factor on 330 kg corner mass) yields a maximum von Mises stress of 5.357 MPa, maximum displacement of 0.000714 mm, and a minimum factor of safety of 51.34 — substantially exceeding the IATF 16949 minimum of 2.0. The stress utilisation ratio of 1.95 % conclusively demonstrates structural over-design.
- Fourth — Topology optimisation: SIMP-based topology optimisation identifies a theoretical minimum element mass of 0.10543 kg, a 77.1 % reduction from the 0.46002 kg baseline. A manufacturable redesign incorporating a rear face pocket and two lightening slots achieves the topology target while remaining fully producible by the existing CNC milling process route.

Engineering Achievements

The report demonstrates the successful integration of three engineering domains — CAD parametric modelling, CAM toolpath generation, and CAE structural analysis — within the unified SolidWorks 2024 environment. The investigation has shown that quantitative engineering metrics drawn from physical principles (cutting speed equations, von Mises stress theory, modified Goodman fatigue criterion) provide superior justification compared to descriptive arguments alone. The application of six recognised manufacturing standards (ISO 2768, ISO 1101, ISO 13849-1, ISO 9001, ISO 1302, IATF 16949) ensures that the design is unambiguously specified, reliably manufacturable, and compliant with the safety expectations of automotive structural components.

Optimisation Outcomes

In aggregate, the proposed manufacturing and design optimisations are projected to deliver per-component cost reduction of 23–33 % (combined material and machining savings), per-line annual energy savings of 600 kWh, per-line annual CO_2 avoidance of 351 kg, and a 25 % reduction in tungsten carbide consumption — directly aligning with United Nations Sustainable Development Goals 9 (Industry, Innovation and Infrastructure), 12 (Responsible Consumption and Production), and 13 (Climate Action), and with Malaysia's National Energy Transition Roadmap 2023.

Recommended Future Work

Three follow-on investigations would extend the engineering value of this study. (i) A dedicated fatigue study should be conducted on the topology-optimised geometry using SolidWorks Simulation Fatigue Module with a sinusoidal load history at 1–10 Hz frequency consistent with automotive suspension durability testing per SAE J1211, to verify infinite-life behaviour at the elevated post-optimisation stress level. (ii) A formal mesh convergence study should be performed on the redesigned bracket — successively halving the global element size until peak-stress convergence falls below 5 % — to validate the accuracy of the FEA prediction in the geometry where the stress utilisation ratio increases from 2 % to 5 %. (iii) A modal analysis is recommended to characterise the natural frequencies of the redesigned bracket and verify a separation margin of ≥ 30 % from the dominant suspension excitation frequencies, eliminating any potential resonance with road-induced vibration inputs.

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